ALL

Introduction

MACRO

The projected effects of urbanization and climate change on urban Heat Island and thermal comfort over the Tehran metropolitan

Link: https://www.sciencedirect.com/science/article/abs/pii/S0048969725015955

“Rapid urbanization and changes in land-use patterns have intensified global environmental challenges in megacities. As cities confront the dual pressures of urbanization and climate change, understanding their individual and combined impacts on the thermal environment and urban resilience remains a critical research priority.Rapid urbanization and changes in land-use patterns have intensified global environmental challenges in megacities. As cities confront the dual pressures of urbanization and climate change, understanding their individual and combined impacts on the thermal environment and urban resilience remains a critical research priority.

The results reveal that although cool roof helps mitigate the effects of urbanization and climate change on temperature rise and thermal comfort by up to 4.37 % and 15 %, respectively, it also exacerbates the reduction in wind speed by 8.58 %, hindering pollutant dispersion. This study shows the need to integrate urbanization and climate change impacts into sustainable urban planning, offering valuable insights for targeted adaptation and mitigation strategies.”

Handbook on Urban Heat Management in the Global South

Link:https://www.worldbank.org/en/topic/urbandevelopment/publication/handbook-on-urban-heat-management-in-the-global-south

“**Extreme urban heat** is one of the biggest challenges facing cities today, especially in the Global South, where rapid urbanization and limited resources make adaptation harder. Rising temperatures disrupt work, schooling, and access to health services. Heat also drives up energy demand for cooling, straining energy systems and increasing emissions. Urban heat islands make cities even hotter—sometimes by up to 10°C—threatening lives, productivity, and economic stability.

The costs of inaction are severe. Heatwaves will become frequent and intense, with deadly impacts on vulnerable groups such as outdoor workers, the elderly, and low-income families. By 2050, the number of urban poor exposed to extreme heat could rise by 700%, with the largest impacts in West Africa and Southeast Asia. Without proactive measures, cities risk worsening inequality, economic losses, and migration pressures. Expanding equitable access to sustainable cooling is essential to protect communities and economies.”

Urbanization is projected to increase local surface temperature by 2100

https://www.nature.com/articles/s43247-025-02947-1

“Future projection of global land surface temperature often emphasizes climate

change while neglecting urbanization. Yet, urbanization-induced warming strongly influences heatwave-related health risks and energy demands. Here, we developed a 1-km resolution global land surface temperature dataset for 2020–2100 at five-year intervals, combing climate change-induced global warming and urbanization-driven local warming, which were estimated using multi-model ensemble projections, and a dynamic regression model linking impervious surface area and local temperature, respectively.

By 2100, urbanization contributes an average local warming of 0.1 °C, with approximately 10–16% of urban areas experience extreme warming exceeding 1 °C. Urban areas remain warmer than the global mean, whereas their warming rates are 0.5–8% lower than the global average under all scenarios. The derived dataset enables improved assessments of urban heat risks assessments and supports climate-resilient urban planning.”

Urban heat in global cities and the role of nature-based solutions in mitigating future climate risks
https://www.researchgate.net/publication/390669600_Urban_heat_in_global_cities_and_the_role_of_nature-based_solutions_in_mitigating_future_climate_risks

Approximately eight billion people are living on Earth today with more than half (55%,
~4.2 billion) living in cities—a proportion predicted to increase to 70% (~6.6. billion) by 2050. As
the human population grows, urban residents will face increasingly extreme temperatures under
future climate change, which will affect human well-being, health, and mortality.

A 1940-2020 spatiotemporal analysis of thermal discomfort days in Southeast Asian countries

“”By comparing each country’s UTCI and MRT records between past (1940–2000) and present (2001–2020), a consistent and significant increase in TDD occurrence frequency was found across all SEA countries. While the mean increase in UTCI was relatively modest, the corresponding rise in TDD likelihood was disproportionately larger, indicating a heightened risk of TDD exposure. A distinct contrast was revealed between continental and maritime regions, where maritime countries showed a relatively smaller rise in absolute values of both indices compared to their continental counterparts. However, both the slopes of the fitted trendlines and the coefficients of determination of the linear fitting were notably larger in maritime countries than continental countries. Such distinction indicates that maritime regions could benefit from the moderating effects of the surrounding oceans as water with higher heat capacity heats up slower than land, but the pronounced rising trends in TDD occurrence frequency suggest a more rapid escalation in heat stress despite the relatively smaller absolute increase in maritime countries.”
“Looking ahead, the projections for 2050 suggest a continuation of warming trends, with Indonesia and Malaysia expected to remain at the forefront of TDD

occurrence, while Singapore emerges with the lowest numbers of TDDs. The projected increase in TDD frequency underscores the urgency of proactive measures to address climate change and its impacts, particularly in vulnerable maritime and low-lying areas. In conclusion, the findings presented in this study provide valuable insights into the evolving climate dynamics of SEA and underscore the need for concerted action to enhance climate resilience and adaptation efforts in the region. By leveraging scientific understanding and data-driven analyses, policymakers and stakeholders can better prepare for the challenges posed by rising temperatures and TDDs in the decades to come."

https://doi.org/10.1088/2515-7620/ad810b

A review of urban resilience to weather and climate extremes
https://link.springer.com/article/10.1007/s44213-025-00063-6

"Urban areas are increasingly vulnerable to extreme weather and climate events, a risk amplified by rapid urbanization and climate change. As hubs of economic and social activities, cities face escalating threats from heatwaves, floods, storms, and air quality degradation, with implications for public health and safety as well as long-term sustainability. This review synthesizes current understanding of urban vulnerability and resilience to climate extremes, emphasizing the interactions among physical infrastructure, socioeconomic conditions, and governance systems that shape a city's capacity to withstand and recover from disruptions. Key barriers to resilience include inadequate infrastructure adaptation, the growth of vulnerable informal settlements, limited financial resources, and fragmented policy implementation across governance levels. Importantly, challenges manifest differently across regions: while Global South cities grapple with rapid expansion, informal settlements, and constrained fiscal and institutional capacity, Global North cities contend with aging, interdependent infrastructure systems, and the political complexities of retrofits and managed retreat. Critical strategies to address these challenges involve advancing climate-resilient infrastructure that integrates engineered systems with nature-based solutions, strengthening governance frameworks that prioritize equity and community engagement, leveraging technological innovations such as smart city systems and predictive analytics, and ensuring equitable access to adaptation measures for marginalized populations. Prioritizing these interlinked elements through coordinated action across multiple scales and sectors is critical for fostering resilient, equitable, and sustainable cities capable of withstanding and adapting to increasingly frequent and severe climate extremes in an uncertain future."

"Cities are increasingly threatened by weather and climate extremes, driven by rapid urbanization and climate change. The complex interplay between these factors necessitates a comprehensive understanding of resilience that encompasses both mitigation and adaptation strategies. Mitigation focuses on reducing greenhouse gas emissions and promoting climate-neutral practices, while adaptation aims to enhance the capacity of urban systems to withstand and recover from climate impacts. Effective climate-resilient adaptation requires integrated planning that addresses multiple hazards rather than concentrating solely on individual risks. Cities must develop and implement climate-neutral policies that combine both mitigation and adaptation strategies, prioritizing emission reductions while enhancing resilience to climate extremes. This multi-

hazard resilience planning approach recognizes that urban areas often face simultaneous or consecutive hazards, necessitating comprehensive strategies to bolster resilience across all urban systems."

Applying recent advances in climate adaptation research to urban heat risk management

"There is unequivocal evidence that anthropogenic climate change is supercharging temperature and precipitation regimes globally. One of the clearest signals of this is seen in current and projected increases in extreme heat, understood as changes in temperature maximums, longer duration heatwaves, and higher night-time temperatures. Extreme heat has substantial impacts on socio-ecological systems through direct impacts on human health and labor productivity, crop yields and water security; and second-order impacts on infrastructure functioning and hazards (e.g., increased fire and drought incidence)."

Cooling the cities: A comprehensive review of urban heat island mitigation strategies in Southeast Asia

"This systematic review synthesizes 27 peer-reviewed studies (2020–2023) on urban heat island (UHI) mitigation in Southeast Asia, focusing on cooling effectiveness and implementation feasibility. In dense cities like Singapore and Kuala Lumpur, green infrastructure—urban forests, green roofs, and street-tree networks—proves most effective, reducing air temperatures by up to 2.6 °C and surface temperatures by 11 °C. Engineered solutions, such as high-albedo coatings and permeable pavements, offer 4–9 °C of surface cooling and added benefits like stormwater control, though tropical humidity, glare, and maintenance pose challenges. Informal settlements demonstrate the value of low-cost, community-led cooling efforts using shade structures and vegetation. Despite these benefits, barriers persist: high costs, limited urban land, and fragmented governance systems hamper the scalability of hybrid green-grey solutions. Evaporation-based strategies are further constrained by complex urban geometries and monsoonal climates. Fewer than 20 % of Southeast Asian cities incorporate UHI mitigation into formal planning frameworks, limiting systemic uptake. To advance climate-resilient urbanism, interdisciplinary collaboration is needed. Future priorities should include long-term monitoring in tropical climates, integrated trials of hybrid interventions, and cost–benefit models that capture health, energy, and flood resilience co-benefits. Embedding these approaches into planning systems, as seen in Bangkok's zoning reforms and Singapore's Green Mark, is key to sustainable adaptation."

Evaluating contributions of urbanization and global climate change to urban land surface temperature change: a case study in Lagos, Nigeria

"This study develops a general method to evaluate the contributions of localized urbanization and global climate change to long-term urban land surface temperature (ULST) change. The method is based on the understanding that long-term annual ULST is controlled by three factors: (1) localized urbanization, (2) global climate change, and (3) interannual climate variation. Then the method removes the interannual climate fluctuations on long-term observed LST time series via linear regression and separates the contributions of urbanization and climate change to the impacts on longterm ULST via urban–rural comparison. The method is applied to Lagos, a fast-growing metropolis in the tropical West Africa,

as an example for reference”

Exploring the impact of urban features on land surface temperature over South Asian cities

“Extensive research has established that urban land cover plays a critical role in modulating land surface temperature (LST), with urban morphology and surface heterogeneity driving its spatial and temporal variations. Studies have also highlighted that the increasing frequency, intensity, and duration of extreme heat events exacerbate urban thermal stress, significantly impacting livability and thermal comfort. This study utilizes high-resolution (70 m) LST data from NASA’s Ecosystem Spaceborne Thermal Radiometer Experiment on Space Station (ECOSTRESS) to investigate the diurnal cycling of LST and its variations across local climate zones (LCZs) in two major South Asian cities, Delhi and Lahore. ECOSTRESS LST images from four distinct heatwave events (2020–2023) were analyzed to quantify intra-LCZ variations and surface urban heat island (SUHI) intensity at different times of the day and night. Results indicate significant diurnal thermal disparities among LCZs, with the highest average SUHI observed in Lahore (3.48°C), while Delhi also exhibited a positive SUHI effect (0.46°C). Findings revealed distinct spatial LST patterns across LCZs, highlighting the role of urban vegetation and impervious surfaces in regulating SUHI intensity. Additionally, urban heat hotspots and coldspots were identified, largely influenced by the characteristics of the underlying LCZ types. The study provides critical insights into the spatiotemporal dynamics of SUHI during extreme heat events, offering a valuable reference for urban planners and policymakers in designing climate-resilient cities and mitigating urban heat stress.”

Financing structural and non-structural extreme heat adaptation measures in Southeast Asian cities: statuses and prospects

“Extreme heat, often called a ‘silent killer,’ poses significant threats, particularly in Southeast Asia, where rapid urbanisation and increasing population density exacerbate vulnerability. Populations in this region face heightened risks of heat-related mortality and illness and diminished efficiency of the energy systems essential for cooling. Additionally, limited financial resources impede cities’ adaptation efforts, establishing finance as a critical factor influencing vulnerability to extreme heat. Despite growing recognition of the necessity for climate adaptation and mitigation financing, as highlighted by the Sendai Framework for Disaster Risk Reduction and the outcomes of the Conferences of the Parties to the United Nations Framework Convention on Climate Change, the adaptation finance gap continues to widen.”

How do disparate urbanization and climate change imprint on urban thermal variations? A comparison between two dynamic cities in Southeast Asia

“This study assessed the combined influences of climate change and urbanization on land surface temperature (LST) and surface urban heat island (SUHI) in the Bangkok metropolitan region (BKK) and Ho Chi Minh City metropolitan area (HCM) during 1990–2020. We found climate change evidence from trends and variations of temperature extremes in both cities throughout the past three decades, especially for the significant upward trends of nighttime temperature in HCM (0.23 – 0.3 °C/decade). These cities have been undergoing rapid urbanization, in which HCM has a higher urbanization rate with an annual growth rate of 8.03%/year

against BKK (4.26%/year). Landscape metrics have rapidly shifted towards urban aggregation and green space fragmentation along with urbanization. A peer comparison shows that BKK holds a greater urban size and LST intensity, while HCM leads in urban growth and SUHI magnitude. These changes jointly contribute to climbing SUHI intensity, especially the dramatic influence of surface changes. Yet, temperature extremes also play an indispensable role in regulating SUHI magnitude, in which the impact in HCM is considerable compared to the effect in BKK. The research findings provide crucial information for both cities – where incorporation of climate change resilience strategies and SUHI mitigation into the master plans for livable cities are recommended."

Impact of Land Use Land Cover Change on Land Surface Temperature in Small and Medium Sized Cities in South Asia: A Remote Sensing Approach

"Rapid urbanization impacts the surrounding urban environment in multiple ways, with the Urban Heat Island (UHI) effect being one of the most critical, driven by concrete surfaces and anthropogenic activities that create a rural–urban temperature difference. Climate change, coupled with these land use changes, increases the vulnerability of city dwellers to extreme heat. South Asia, home to millions of urban residents, faces heatwaves that are rising in both intensity and frequency. "

"Heatwaves are increasingly impacting the life and livelihood of South Asian cities, with India alone witnessing a 55% rise in heat-related deaths between 2000–2004 and 2017–2021, highlighting the growing vulnerability of urban populations to extreme heat events (Akhtar, 2024). The increasing built-up area and reduced natural surfaces heavily contribute to the UHI effect. Nine cities in five countries - Bangladesh (Rajshahi, Chattogram), India (Ahmedabad, Bhubaneswar, Kanpur), Pakistan (Gwadar, Quetta), Nepal (Kathmandu), and Afghanistan (Kabul) - were studied to examine the LULC change pattern and its impact on LST. Using satellite remote sensing and GIS techniques, this study clearly shows that all nine cities in South Asia have an increasing built-up area between 2003-2023, which is taking over natural surfaces like vegetation and water. The mean annual LST has increased in all the cities, with some showing higher increase and others moderate increase. The correlation analysis indicates that despite microclimatic differences, geographic location (continental or coastal), or landform (wet or dry), the built-up area has the highest influence on LST. Vegetation has a medium cooling impact in cities, and water bodies contribute very little to the cooling effect because of the very limited presence of water in these cities. These findings reveal that unchecked urban growth is steadily intensifying heat stress across South Asia, regardless of geography or microclimate. Without proactive urban planning and investment in green infrastructure, the risks to public health, productivity, and urban sustainability will continue to escalate. Strengthening climate-resilient policies and prioritizing nature-based solutions can therefore play a crucial role in shaping cooler, healthier, and more livable cities for the future."

Impact of urbanization on exposure to extreme warming in megacities: Heliyon

"Cities warm up due to two main factors: global climate change and urbanization-induced warming (so-called, urban heat island effect). In the projection of future climate, coarse-resolution global climate models are not suitable for looking into

the heterogeneous urban surface and their changes. On the other hand, regional climate models, which are capable of looking into cities in detail, have never been used to investigate the global urban climate. Here we show that urbanization significantly increases exposure to extreme warming for megacity residents. We reflect urbanization between the 2010s and the 2050s into our model by considering the spatiotemporal change in urban surface (buildings and anthropogenic heat emissions) induced by urban population and economic growth. We found that in the 2050s, under the worst-case scenario, 78 percent of megacity residents will be exposed to 2.5°C warming, much higher than the projection of 65 percent when urban warming is left out. Our results highlight the importance of accounting for local urbanization in future global urban climate projection."

Integrating Satellite-Based and Participatory Spatial Analyses for Local, Contextualised, and Nuanced Extreme Heat Hotspot Mapping in Overheated Urban Southeast Asia

Rapid urbanisation in Southeast Asia intensifies extreme-heat exposure, especially where high population density, extensive impervious surfaces, and limited green cover coincide. This study combines multi-decadal satellite-derived land surface temperature (LST) mapping with spatial regression and community participatory mapping to locate community-level heat clusters of extreme heat in Metro Manila and the Bangkok Metropolitan Area.
This study demonstrates the value of integrating satellite-based LST mapping, spatial regression, and community participatory mapping to characterise extreme-heat exposure in two Southeast Asian megacities. Satellite LST identified spatial clusters of elevated surface temperatures across Metro Manila and the Bangkok Metropolitan Area. Spatial regression linked these clusters to a set of built-environment, infrastructural, natural-feature, and socioeconomic covariates. Crucially, participatory mapping supplied place-based knowledge that both complemented and contextualised the remotely sensed patterns—revealing everyday exposure pathways, locally salient hotspots, and coping practises that cannot be derived from LST alone.
The combined evidence reinforces the centrality of green infrastructure and shading in mitigating urban heat. It indicates meaningful policy levers—ranging from targeted greening and shade provision in mapped hotspots to land use and transport decisions that reduce heat generation and improve accessibility. The policy-relevant indicators (e.g., efficiency ratios and greenspace accessibility) yield actionable metrics that urban planners and local policymakers can use to prioritise interventions in resource-constrained settings.

Remote sensing-derived land surface temperature trends over South Asia

The spatiotemporal variations in temperature (global, regional, and local scales) and related consequences of climate change are threatening human existence, ecological communities, and socioeconomic development across the world Over the past decades, the global temperature is mostly increasing at a notable rate, which plunders the living planet Earth by disrupting the normal nature balance and forcing extreme changing weather patterns. Understanding the principal driving

factors of warming/cooling trends is the cornerstone for the better implementation of appropriate adaptation and mitigation policies under the temperature changing scenario for a sustainable environment. However, the detailed perception, quantification, and interpretation of the temperature variations, their driving factors, and related consequences have not been done yet for many regions of the world, including the South Asia region.

Spatio-temporal Distribution of Heat Index and Land Cover Change in Tropical Cities of Southeast Asia

Southeast Asian countries lie surrounding the equator line, which makes them have a tropical climate, and periodically suffer extreme heat events due to the combination of high temperature and high humidity. Jakarta, Bangkok, Kuala Lumpur, and Manila are the four largest capital cities in Southeast Asia that we attempted to identify those patterns of the maximum heat index and land cover change during 2001 – 2016.

Spatiotemporal Patterns of Land Surface Temperature, Urban Heat Island, and Potential Heat Stress Risk Areas Assessment for Tropical Inland City and Coastal City in Thailand

Global warming and rapid urban growth, compounded by the urban heat island (UHI) effect, are posing substantial challenges for urban populations and ecological well-being due to increased heat stress risks.

The state and future of extreme heat studies in Southeast Asian megacities: risk, impacts and adaptation strategies in a warming world

Southeast Asian cities are increasingly affected by heat-related phenomena and various climate-related disasters; however, research on urban heat in this region remains limited compared to other areas. This paper employs bibliometric and thematic analyses to investigate studies on extreme heat in megaurban areas of Southeast Asia, focusing on the risks and impacts faced by vulnerable populations and their adaptation and mitigation strategies. The bibliometric analysis visualises the research landscape, identifying key clusters and highlighting prevalent themes and gaps. It reveals a predominant emphasis on characterising extreme heat and analysing urban temperature variations through satellite and meteorological data, underscoring a significant lack of research on the socio-economic factors affecting at-risk communities. The thematic analysis further examines how existing studies address these socio-economic vulnerabilities and evaluates the adaptation strategies employed, particularly concerning land use changes driven by population growth. Our findings indicate that, while studies address urban heat in Southeast Asia, their quantity is relatively small compared to the extensive research focused on other regions. Strategies to mitigate the effects of extreme heat on mental and social well-being emphasise the importance of green infrastructure and public spaces. There is also a pressing need to enhance urban planning and design to ensure adaptation measures include at-risk, lower-income communities. Understanding the complexities of the risks and impacts of extreme heat on urban populations is crucial for developing effective, context-specific adaptation strategies that prioritise the needs of vulnerable populations and

promote equitable, sustainable urban development

Thermal changes along the urban-rural continuums in Southeast Asia
“ Southeast Asia’s (SEA) rapid urbanization has intensified urban heat effects. While previous studies address urban warming at specific time points, the dynamic relationship between land cover and land use (LCLU) changes and land surface temperature (LST) along urban-rural continuums remains understudied.” “This study investigated the effects of urbanization on urban thermal variations across URCs in SEA over the past two decades. Our findings revealed distinct LST change patterns across varying city sizes, with larger cities experiencing broader and more intense urban warming that extends into rural areas. While urbanization significantly contributed to urban warming in SEA, our results show that NDVI had a stronger influence on LST trends. The impact from green vegetation biomass reflects its cooling effect via evapotranspiration and shading, and the warming effect from vegetation loss due to urban expansion. Notably, some urban areas experienced a decline in heat due to increased tree crown closure over time. We recommend urban development strategies that balance urbanization and warming, such as promoting growth in smaller surrounding cities and encouraging outward expansion from densely populated urban centers into rural areas.”

Thermal intensification and urban land cover change in Southeast Asian capitals: over three decades of satellite observations and trend analysis

In conclusion, this multi-decadal analysis of eleven Southeast Asian capital cities reveals a persistent intensification of thermal effects, closely associated with rapid urban expansion between 1990 and 2024. Increases in impervious surface areas, combined with the decline of vegetation and water bodies, have contributed to elevated LSTs and accentuated thermal contrasts between urban cores and their peri-urban fringes. Rising IUHII values indicate increasing thermal discomfort and a heightened exposure to heat-related risks in rapidly urbanising tropical environments. By providing spatially consistent, long-term observations across an underrepresented equatorial region, this study helps to address a notable geographic and climatic gap in the global UHI literature, which has largely centred on temperate, mid-latitude cities. The findings provide an empirical basis for developing regional heat mitigation strategies and urban design approaches that support thermal resilience in fast-growing metropolitan centre

Urban Thermal Environment Under Urban Expansion and Climate Change: A Regional Perspective from Southeast Asian Big Cities

“Southeast Asia (SEA) is a hotspot of rapid urbanisation and climate change, which is supposed to influence millions of people. They altogether exacerbate urban thermal environments and amplify urban heat islands (UHIs). The chapter provides a comprehensive assessment of urban warming trends in SEA big cities over the past thirty years in a connection with urban expansion and climate changes; of which climate change is represented by temperature extremes as an additional element deteriorating UHIs. It reveals a significant warming trend in nighttime extreme indices across the cities against daytime indices. Manila and Kula Lumpur have confronted the most prominent temperature variations. There

were drastic transformations in urban areas, urbanisation patterns, and landscape characteristics during urbanisation. The fastest-growing cities during this period are Jakarta, Bangkok, and Ho Chi Minh City. The degradation of the urban thermal environments was also confirmed via spatiotemporal escalations of land surface temperature (LST) and surface UHI (SUHI) over time. The aggravation in urban thermal environments was mainly driven by urbanisation-related elements and temperature extremes as a background climate factor. The changes in surface characteristics are the main driving factors across the cities. Yet, the stimulations of climate on thermal environment degradation are uneven among the cities. Ho Chi Minh City, Jakarta, and Manila are the most vulnerable cities due to combined impacts rather than others. It verifies the integrated impacts of climate change and urbanisation on urban microclimate change, it therefore should have long-term strategies to mitigate the adverse effects of SUHI. The general principle is to reduce horizontal urban expansion, ensure urban spatial planning and urban green spaces, and mainstream climate change adaptation into urban planning."

Widespread heat stress will become the norm in a warming Southeast Asia

Southeast Asia faces growing risks from extreme heat under climate change, yet existing assessments do not fully account for physiological impacts and population-level exposure, particularly at resolutions relevant for planning.

Our findings extend upon earlier regional studies which demonstrated that heatwaves in Southeast Asia would become more extreme and frequent with climate change, based on temperature indices1,11–13. While these studies illustrate there will be increasing risk from extreme heatwaves in a warming climate, the use of bioclimatic indices in this study help to contextualise the impacts on health and labour productivity

Will population exposure to heat extremes intensify over Southeast Asia in a warmer world?

Temperature extremes have increased during the past several decades and are expected to intensify under current rapid global warming over Southeast Asia (SEA). Exposure to rising temperatures in highly vulnerable regions affects populations, ecosystems, and other elements that may suffer potential losses.

Heat extremes have imposed huge social and ecological stresses in the last few decades. If global warming continues, heat extremes are expected to be more frequent and will exert a negative impact on humans and ecosystems. It is critical for researchers to quantify the potential risk of heat extremes to populations. In this study, we have evaluated the responses of heat extremes at different GWLs against the present day (1985–2014) and the corresponding impacts on population exposure over SEA.

PHILIPPINES

Understanding Impacts of Excess Heat in Filipino cities

The Project Development Objective is to empower urban Philippines Local Government Units (LGUs) to take preventative and real-time action to mitigate the

impacts of heatwaves through provision of more timely and granular data products that reveal localized occurrences of heatwaves and their impacts on human health and urban economies.
Urban heatwaves have cascading impacts on the economies and well-being of people, including heat-induced hospitalizations, power-grid failure, declines in labor productivity, shortened school days, and others. Unless the impacts of heatwaves are mitigated, Filipino cities, with their high population density, will struggle to meet their full economic potential and ensure the wellbeing of their people.
Until now, cities have been challenged to address heatwaves, because the data and expertise needed to mitigate impacts are typically scattered across many departments. To overcome this barrier, as part of the World Bank's Technical Deep Dive on Urban Heat (April 24-28, 2023), Filipino authorities have requested support on better measurement of urban heatwaves and their impacts. To meet this demand-driven request, the project team would facilitate access to heat risk and vulnerability information for the most populous cities in the three regions of the Philippines. Through this public data good, urban governments will be able to identify investments and policy interventions that build resilience at a community level and support scaling up the resilience of the city. The knowledge product will be validated through local data collection and stakeholder participation to ensure inclusive design and participation.
In the Philippines, more than 40 percent of the population lives in cities, and many of these cities are at high risk of experiencing heatwaves.Its position as part of Southeast Asia, with high temperatures and humidity already being the norm, and the 7641 islands that make up the country are especially susceptible to climate change. Challenges in exposition, vulnerability, and adaptability to climate made the Philippines become the country with highest extreme weather disaster-related risk in the world according to the 2022 World Risk Report. Additionally, rapid urbanization is expected to continue as "drought and low farm productivity" drive migrants into cities. The consequences of heatwaves to these residents will be profound, and call for new methods to better understand their behaviors and propose better mitigation strategies. This intersects with the high inequality found in Filipino cities: in the first semester of 2021, driven by the Covid-19 pandemic, 23.7% of the country's overall population was considered poor. Studies have shown strong correlation between poverty and vulnerability to heatwaves, and low-income housing has poor thermal performance and high thermal discomfort levels across the globe. Even without taking into consideration socioeconomic status, a vast majority of Filipinos experience heat stress severely (as self-reported), and this increases with population density.
The serious consequences are already being felt, with recent months showcasing extreme temperatures. In April 2023, for example, dangerous heat was observed throughout the country.
In this notebook, we visualize the areas within Filipino cities that are adversely impacted by heat using multiple data sources, and try to understand some of the aspects that have led to this upending crisis.

The effect of urbanization on temperature indices in the Philippines

The Urban Heat Island (UHI) effect or the observed warming in urban areas relative to the surrounding rural areas, is one of the prominent impacts of

urbanization and has been associated with land-use changes, increased retention of solar heat by some building materials with low albedo and high heat capacities as well as the increased anthropogenic heat emissions (e.g., transportation, industry, and heating from the usage of air conditioning units)
The Philippines experiences several climatic hazards and environmental challenges as a result of climate change and urbanization. As such, it is essential to obtain an in-depth understanding of extreme climate change and the potential impact of urbanization on observed climatic change

Long-term trends and extremes in observed daily precipitation and near surface air temperature in the Philippines for the period 1951–2010

Cinco *et al*. (2014) analysed temperature trends in the Philippines, suggesting that days and nights were becoming warmer. However, their findings were influenced by several factors, such as urbanization, which plays an important role in observed temperature trends.

When the city heats up: mapping urban heat risks through environmental and socioeconomic factors in Quezon City, Philippines

While urbanization drives economic growth and development, it also intensifies urban heat, worsened by climate change and urban heat island effects, which continue to threaten public health, livability, and urban resilience in cities. This study develops and maps a localized heat risk index (HRI) for Quezon City, Philippines, integrating environmental exposure and socioeconomic vulnerability variables through a weighted framework derived using analytic hierarchy process based on expert input. Spatial analysis revealed that 81% of barangays, accounting for 70% of the city's population, fall under high-risk HRI classification (0.61–0.80), with the highest mean HRI recorded in Barangay Manresa (0.731). High-risk areas are concentrated in the southern and southwestern zones of the city, characterized by dense urbanization, limited vegetation, and high surface temperatures. Intraurban heat islets, covering 31% of the study area, strongly overlap with high-risk HRI barangays, emphasizing the compounded effects of environmental and socioeconomic factors on heat risk. The study provides insights to complement the Quezon City Enhanced Local Climate Change Action Plan 2021–2050 by identifying priority areas for implementing nature-based solutions, such as expanding green corridors and enhancing urban biodiversity. These findings highlight the critical role of integrating vegetation, reflective surfaces, and reduced built-up densities in mitigating heat risks and improving urban resilience. The HRI framework offers a replicable model for assessing urban heat risks, supporting evidence-based policymaking, and enhancing adaptive capacity in other rapidly urbanizing cities.

Assessing the Effects and Dangers of Climate Change on the Environment and Society of the Philippines

The Philippines is one of the most susceptible nations to climate change's effects. It is anticipated that the frequency and severity of extreme weather events such as

typhoons, floods, and droughts will increase over the next few decades. This thesis seeks to evaluate the environmental and social hazards posed by climate change in the Philippines. Examining the current and projected effects of global warming on critical sectors such as agriculture, coastal zones, and human health. The research methodology will include a literature review, data analysis, and case studies. This study's findings will provide crucial insights into the resiliency and vulnerability of the Philippine community to the effects of climate change. This research will contribute to the development of adaptation and mitigation strategies that will assist the nation in building resilience and reducing the risks associated with global warming. This paper will be of interest to policymakers, academics, and practitioners working in the field of climate change and sustainable developme

This literature review assesses the effects and hazards of climate change on the environment and society of the Philippines, drawing from numerous sources. The findings disclose significant and extensive repercussions, including effects on biodiversity, ecosystems, water resources, agriculture, food security, coastal and marine resources, public health, human settlements, infrastructure, economy, displacement, and migration. Geophysical, socioeconomic, and institutional factors influence the Philippines' susceptibility to climate change, necessitating comprehensive and integrated mitigation and adaptation strategies.

Assessment of Intra-Urban Heat Island in a Densely Populated City Using Remote Sensing: A Case Study for Manila City

Changes in the environment occur in cities due to increased urbanization and population growth. Sustainable Development Goal (SDG) 11 is intrinsically linked to the environment, one facet of which is the need for universal access to secure, inclusive, and accessible green and public places. As urban heat islands (UHI) have the potential to negatively influence cities and their residents, existing resources and data must be used to identify and quantify these effects. To address this, we present the use of satellite-derived (2013–2022) and meteorological data (2014–2020) to assess intra-urban heat islands in Manila City, Philippines. The assessment includes (a) understanding the temporal variability of air temperature measurements and outdoor thermal comfort based on meteorological data, (b) comparative and correlative analysis between common Land-Use Land Cover indicators (Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI) and Normalized Difference Built-up Index (NDBI)) and Land Surface Temperature (LST), (c) spatial and temporal analysis of LST using spatial statistics techniques, and (d) generation of an intra-urban heat island (IUHI) map with a recommended class of action using a suitability analysis model. Finally, the areas that need intervention are compared to the affected population, and suggestions to enhance the thermal characteristics of the city and mitigate the effects of UHI are established.

This study presents the use of satellite-derived data and meteorological data to assess the presence of an intra-urban heat island in Manila City, Philippines. To address SDG 11 and provide better insights to make cities and human settlements inclusive, safe resilient, and sustainable in terms of UHI, different assessment methods were used and established. The assessment includes (a) understanding the temporal variability of air temperature measurements and outdoor thermal comfort based on meteorological data, (b) comparative and correlative analysis

between common LULC indicators (NDVI, NDBI, and NDWI) to LST, (c) spatial and temporal analysis of LST using spatial statistics techniques, and (d) generation of an intra-urban heat island (IUHI) map with a recommended class of action using a suitability analysis model. Finally, the areas that need intervention are compared to the affected population, and suggestions to enhance the thermal characteristics of the city and mitigate the effects of UHI were established. Results show that there exists a clear difference between cold and warm areas within Manila City. Overall, residential areas, asphalted and concrete roads and walkways, and some commercial establishments and buildings exhibit higher surface temperatures compared to areas with vegetation and near bodies of water. Based on the results, mitigation strategies applicable to Manila City were proposed to improve the areas which need intervention.

Land-use/land cover change and land surface temperature in Metropolitan Manila, Philippines using landsat imagery

Remote sensing technology has been widely used to address the issue of urbanization for monitoring. In Metropolitan Manila, Philippines, there is a decline in vegetation over the last decade. This leads to the excess heat in the surface temperature and a phenomenon known as Urban Heat Island. In this study, the land surface temperature (LST), land-use/land cover changes (LULC), normalized difference vegetation index (NDVI) and normalized difference built-up index (NDBI) in 2001 and 2019 were retrieved using Landsat 7 Enhanced Thematic Mapper + (ETM +). Our results showed the following: (i) an increase in the mean temperature of 4.02 °C from 2001 to 2019 in MM, (ii) rise in NDBI and a decline in NDVI values is correlated with an increase in LST over the last years in MM; (iii) LULC maps affirmed the highest occupancy of built-up areas caused by possible urban sprawl explaining the increase in warming in MM. This study provides added evidence that urbanization activities in Metro Manila leads to a warming effect that drives discomfort to many average Filipinos. Hence, environmental policies that would implement proper urban design utilizing more green spaces is recommended for local government units in MM to support.

Modeling the influence of land cover dynamics on spatio-temporal variations in land surface temperature in Cagayan de Oro River basin, Mindanao, Philippines

The onset of agricultural intensification and urban expansion in Cagayan de Oro River basin has been continuously changing its landscape over the years. While it undergoes modifications its land cover, its associated impact on the land surface temperature over the years has not been evaluated. This study aims to model the spatio-temporal dynamics of land cover (2003–2020) and determine its influence on the land surface temperature variations in Cagayan de Oro River basin, Mindanao, Philippines from 2003 to 2020 using geographical information systems (GIS) and remote sensing techniques. Significant changes were observed in annual and perennial croplands which experienced substantial increases and is preceded by built-up areas. Considerable decreases were however recorded in shrubland, grassland, mangroves, and open forest. Between 2003 and 2020, the mean land surface temperature in Cagayan de Oro River basin increased by 5.48 °C. The results indicate that built-up areas and bareland consistently recorded the highest mean LST values, while water bodies, mangroves, and forest

cover had the lowest temperatures. This study promotes evidence-based decision-making for sustainable land use planning and management to mitigate the increasing land surface temperature in Cagayan de Oro River basin.

Potential effect of urbanization on extreme heat events in Metro Manila Philippines using WRF-UCM

The impacts of extreme heat events in urban areas are essential, particularly in highly dense megacities such as Metro Manila (i.e. 21,765 people per sq. km as of 2020), where health-related risks are high (HHRI = 0.6-1.0). The health-related risks are associated with extreme heat events, such as heat stroke, dehydration, and exacerbation of existing health conditions. These risks are particularly concerning in densely populated urban areas like Metro Manila, where the urban heat island effect can further amplify the heat stress. WRF-UCM simulations are conducted to analyze five extreme heat events, incorporating urban land use and AH to represent 2015 conditions in Metro Manila. In summary, the simulations using URBAN land cover capture 2m temperature well, especially during nighttime. While the various PBL and radiation scheme options showed similar results, especially in correlation with observation, the ACM2 PBL scheme had the best performance in error metrics. This skill could be attributed to ACM2's ability to capture nighttime temperature best and simulate higher temperature values well. Furthermore, the RRTMG radiation showed better distribution and New Goddard radiation reduced mean biases.

Simulation of Urban Heat Island during a High-Heat Event Using WRF Urban Canopy Models: A Case Study for Metro Manila

This present study aims to determine the performance of using the Weather Research and Forecasting (WRF) Model, coupled with the urban canopy models (UCMs), in simulating the 2 m air temperature and 2 m relative humidity in Metro Manila. The simulation was performed during a high heat event on 22–29 April 2018, which coincided with the dry season in the Philippines. The four urban canopy model options that were used in this study include, the bulk (no urban), SLUCM, BEP, and BEM. The results of the simulations were compared with the hourly observations from three weather stations over Metro Manila from the National Oceanic and Atmospheric Administration Integrated Surface Dataset (ISD) and one agrometeorological station in Naic, Cavite. After model validation, the urban heat island (UHI) was then characterized to determine the spatial-temporal variations in the cities of Metro Manila. Statistical results show that the WRF simulation for 2 m air temperature agrees with measurements with an RMSE of <3.0 °C, mean bias error of <2.0 °C, and index of agreement of >0.80. WRF simulation for relative humidity still presents a challenge where simulation errors are higher than the acceptable range. The addition of UCMs does not necessarily improve the simulation for 2 m air temperature, while the use of BEP improved the 2 m relative humidity simulation. The results suggest the importance of using actual urban morphology values in WRF to accurately simulate near-surface variables. On the other hand, WRF simulation shows the presence of urban heat islands, notably in the northwest and central area of Metro Manila during daytime, extending throughout Metro Manila during nighttime. Lower air temperature was consistently observed in areas near Laguna Lake, while higher air temperature due to stagnant winds was observed in the northwest area of Metro Manila. High heat

index was also observed throughout Metro Manila from daytime until nighttime, especially in areas near bodies of water like Manila Bay and Laguna Lake due to high humidity.

UHII simulation shows hotspots in Metro Manila, especially during nighttime, indicating the presence of urban heat island. U HII values reached to a maximum of 3 °C higher during nighttime and 8 °C higher during daytime in Metro Manila. The presence of water bodies significantly reduces the air temperature of surrounding areas. This is observed where SMD has a lower air temperature compared to NMD and EMD, since it is located near Laguna Lake. Although, during nighttime, SMD is observed to have a higher air temperature due to Laguna Lake acting as a heat source during the hot season. Although air temperature is relatively lower in areas near water bodies, the heat index was observed to be high in these areas due to higher relative humidity. Generally, Metro Manila experiences thermal discomfort throughout the day until nighttime. The high HI at nighttime posits a concern especially for the physiologically vulnerable population who does not have any access to air conditioning facilities, which is relevant to Metro Manila due to the possible effects of heat stress.

SURFACE URBAN HEAT ISLANDS AND RELATED HEALTH RISK IN THE PHILIPPINES: A GEOSPATIAL ASSESSMENT USING MODIS DATA

The study assessed the heat-health risk of the Philippines during 2020 using the MODIS derived Urban Heat Island (UHI) and gridded population data. The study found out that the urban areas experienced more intense heat than the rural areas. It also revealed that the UHI is more intense during March, April and Month or hot dry season. People living in Metro Manila are more exposed and vulnerable to the heat risk. Poverty is one reason that puts people exposed and vulnerable to the heat risk. The output of this study can aid decision-makers especially the urban planners.

Urban Heat Island Effect In The Megacities Of The Philippines

The three megacities namely Cebu City, Davao City and Manila City are considered as highly urbanized cities in the Philippines. The migration of the rural dwellers to these cities are the cause of fast urbanization. To meet the demand for housing, better roads and economic growth, the use of land was shifted from natural ecosystems (such as vegetation, parks, forest) into artificial environments (such as high-rise commercial and residential buildings). These changes in both population densities and land uses greatly alter the weather and climate of these areas.
Studies show urbanization greatly affects local weather, particularly temperature and rainfall patterns of the urban and nearby rural areas. The development of Urban Heat Island (UHI) effect is also documented as this brings discomfort and health related issues among urban dwellers. Using the Diurnal Temperature Range (DTR) values of the selected cities, the researchers investigated the development of UHI. The meteorological parameters of the three cities were generated by the Weather Research and Forecasting Model (WRF).
Results showed that there are more hot days during the hot and wet period than hot and dry periods as revealed by the values of Tmax. Manila was warmer than areas under study during the hot and dry period. Early morning (Tmin , 5-AM)

temperature of Manila is hotter with mean value of 27.3°C. In general, among the three cities Davao had the coolest morning temperature by approximately 1°C – 3°C. The differences of DTR are ranging from 0.15°C to 1°C. The predominance of UHI is observed during the dry and wet period and weak during the cool and dry period. Overall, UHI development occurred for the entire period of study.

DAVAO

Urban heat index in Davao City 'above normal'

The urban heat index in Davao City is already "above normal," Davao City-based environment group Interfacing Development Interventions for Sustainability (IDIS) Inc. said. Manalo said the increasing heat index is attributed to the global warming crisis. He added that rising local temperature is also associated with the influx of population and rapid development of more urbanized areas. He said the city's fully urbanized areas are projected to increase from 4,031 hectares in 2015 to 16,398 hectares in 2045.
Manalo said the key to addressing the rise in urban heat index includes nature-based solutions and low-impact development. "Several government plans and policies have already been put in place, such as the protection of urban trees and remaining forest patches and tree-line canopy avenues, increase of parks, green spaces, landscaping and permeable surfaces," he said. With the temperature rise, he said the implementation of these programs is urgently needed. He said more public covered walkways and waiting sheds must be incorporated into the infrastructure development plans.

URBAN EFFECTS ON LAND SURFACE TEMPERATURE IN DAVAO CITY, PHILIPPINES

This study produced spatiotemporal hot and cold spot occurrence maps for Davao City for the period 1994-2019 using land surface temperature (LST) images. Urban heat is theorized to have been affected by some, if not all, of the following impact factors: air pollutant concentrations/particulate matter (PM10), vegetation "abundance" (using EVI), building "density" (NDBI), albedo, topography, and population density. A mobile traverse sampling was performed in the morning and afternoon of 15 April 2019 to measure PM10 in the city's identified hot spots. The remaining factors were generated from imagery (i.e., Landsat 8, Synthetic Aperture Radar) and obtained from the Philippine Statistics Authority. These factors were analyzed against the LST which was obtained through Project GUHeat's methodology. The relationships between the factors and LST were studied through multiple and quantile regression models (MRM & QRM). Results showed that variable PM10 does not have any significance in the MRM. Meanwhile, QRM were fitted to different quantile values, namely: 10th , 25th , 50th, 75th, and 90th . It is only at the 90th quantile where all the independent variables are good predictors for the LST. Albedo is the most important predictor for the LST at 10th quantile whereas Elev for the 25th quantile. However, when LST is at the 50 th, 75th , and 90th quantiles NDBI is the most significant variable at predicting LST. Reliable spatiotemporal assessment and modelling of surface temperature are essential for urban planning and management to formulate sustainable strategies

for the welfare of people and environment.

This study investigated the spatiotemporal distribution of LST and its quantitative relationships with atmospheric, demographic, and remote sensing-derived parameters in Davao City. The results revealed that occupants of the main urban center (Districts of Poblacion, Talomo, Agdao, and Buhangin) and residents from coastal barangays of Bunawan and Toril Districts experienced the warmest temperatures in Davao City for the period 1994-2019. Multiple regression analysis showed that the air pollutant variable (PM10) exhibited no significance, hence was dropped in the model. Albedo has greater impact in the MRM followed by NDBI, Elev, Pop_Dens, and EVI. Quantile regression analysis further revealed that at the 90th quantile (31.87 °C – 33.42 °C) where all impact factors are good predictors, the presence of buildings in the city (NDBI) showed the biggest impact for the LST.

Spatiotemporal Analysis Of Urban Land Surface Temperature (Ulst) In Davao City

Urban expansion has significantly altered the spatial configuration of green spaces, intensifying the urban heat island (UHI) effect in rapidly growing cities like Davao City. This study investigated the spatiotemporal dynamics of urban land surface temperature (ULST) from 2015 to 2024 across the city's three congressional districts. Using Landsat 9 satellite imagery and geospatial processing in QGIS, annual land surface temperature (LST) trends were derived and analyzed. Results revealed a consistent increase in both minimum and maximum LST values across all districts, with 2024 recording peak temperatures nearing or exceeding 38°C. The most pronounced warming occurred in District I, particularly in highly urbanized barangays such as 1-A, 7-A, and 11-B. Spatial analysis confirmed that densely built-up, vegetation-deficient zones exhibited the highest thermal intensities, while greener, less developed areas maintained cooler surface temperatures. These findings demonstrate the strong inverse relationship between vegetation cover and surface temperature and underscore the ecological implications of urban land use change. Aligned with the Sustainable Development Goals (SDGs), this study calls for the urgent integration of green infrastructure, climate-adaptive planning, and ecological restoration into the city's development agenda. This research contributes critical empirical data to support sustainable urban strategies and enhance climate resilience in Davao City.

GIS-Based Spatio-Temporal Analysis of Vegetation Dynamics, Urban Expansion, and Land Surface Temperature in Davao City

The Earth's surface and atmosphere have been substantially modified by the growth of anthropogenic activities and urbanization. Changes in the composition and operation of the urban ecological system will affect the health of city dwellers by increasing the risk of heat exposure, increasing the risk of violence, illness, and mortality, and reducing labor productivity. Davao City is currently undergoing rapid urbanization, and no comprehensive study has examined the effects of the changes on vegetation indices, urban expansion, traffic density, and land surface temperature, which are likely to affect the area. This study analyzes the spatio-temporal change in vegetation indices, built-up expansion, traffic density, and land surface temperature from 2018 to 2022 using Landsat 8 imagery and GIS tools, particularly QGIS. The results show a moderate negative relationship between

NDBI and NDVI (r = –0.4313), a strong positive relationship between NDBI and LST (r = 0.6216), and a weak negative relationship between NDVI and LST (r =–0.2495). Results highlight the interconnected influence of urbanization, vegetation dynamics, and thermal conditions within the study area, underscoring the need to integrate green infrastructure, particularly along major road networks, to mitigate rising land surface temperatures.

Thus, the study reveals that from 2018-2022; in terms of the vegetation index the moderate vegetation holds the highest area coverage from the aforementioned years this signifies that in terms of vegetation agricultural areas holds the highest coverage in the study area, there are timesthat the total area fluctuates due to the fact that during the capture of data in the landsat 8 coincides the date of pre or post-harvest of the farmers, second the built up index also fluctuates from 2018-2022 this variations due to the changes in the land cover types in where some changes may due to the effect of pandemic which hinders the continues expansion and the practices made by humans such a planting. Third, land surface temperature increases in built-up areas as built-up expansion occurs, whereas very low land surface temperatures are observed in areas with thick, healthy vegetation, which occupies a small land cover area. Fourth, in terms of traffic density, the barangay of Matina and the Davao City International Airport have the highest annual average traffic density across the five major roads. This suggests that almost all people traveled from the south to the north area. This is due to work-related reasons, schools, hospitals, malls, and the Davao airport. The correlation between NDBI and NDVI is negative, indicating that as built-up areas increase, vegetation decreases due to the conversion of vegetated areas to urban areas for infrastructure expansion. Conversely, the NDBI and LST presents a strong positive relationship which indicates that highly built-up areas hold the highest land surface temperature, Lastly, the NDVI and LST present a weak negative correlation emphasizes that areas with highest vegetation cover generally has the lowest land surface temperature. Overall, the results highlight the interconnected influence of urbanization, transportation activities, vegetation dynamics, and thermal conditions within the study area.

This study recommends Local Planners and Policymakers: To strengthen the integration of green infrastructure, particularly along major road networks, to help mitigate increasing land surface temperatures. The establishment of roadside tree planting, urban green buffers, and expanded green spaces is encouraged to reduce heat accumulation in areas with high traffic volume. In addition, land-use planning should be improved to better coordinate transportation development with urban expansion, minimizing fragmented landscapes and uncontrolled surface changes. Regular monitoring using remote sensing indicators such as NDVI, NDBI, and LST is also recommended to track environmental changes and guide evidence-based decision-making.

EXISTING METHODS GAP

Recent advances on machine learning techniques for urban heat island applications: a review and new horizons

Urban Heat Islands (UHIs) pose a significant global urban challenge, exacerbating heat stress, increasing energy demand, and negatively impacting public health.

This review critically analyzes the application of machine learning (ML) strategies for UHI mitigation through an integrated lens encompassing sensing, prediction, optimization, control, and adaptive management. This review starts with a comprehensive evaluation of various data acquisition techniques, such as remote sensing, mobile surveys, and ground-based sensor networks, along with their respective strengths and limitations. Subsequently, the review explores advanced data processing methodologies leveraging ML algorithms for the analysis and interpretation of complex UHI datasets, enabling accurate forecasting and timely interventions. ML-driven prediction and forecasting techniques for UHI are then presented, underscoring the importance of precise and timely predictions for effective mitigation. Further investigation delves into the optimization of UHI mitigation strategies, examining how ML can enhance the effectiveness of approaches such as green infrastructure, cool materials, urban water bodies, and urban planning and design. Finally, the integration of ML insights into flexible adaptation strategies and urban planning processes is discussed, highlighting the necessity for long-term, climate-responsive urban development. The review concludes by assessing the transformative potential and inherent limitations of ML approaches in this domain, outlining critical challenges and promising future research directions for advancing UHI mitigation within rapidly evolving urban environments and under changing climate conditions.

This review has systematically examined the burgeoning application of ML methodologies within the domain of UHI research. By synthesizing insights across diverse data acquisition modalities, sophisticated processing pipelines, and cutting-edge ML models, this work elucidates the significant potential of these approaches to enhance the understanding of complex urban thermal dynamics and inform the development of more efficacious mitigation strategies. The integration of satellite-derived observations, mobile in-situ measurements, and dense sensor networks, coupled with advanced data fusion and analytics, has underscored the transformative capacity of ML methodologies in accurately forecasting UHI patterns and optimizing targeted interventions.
Despite these considerable advancements, several critical challenges warrant careful consideration. The robustness and reliability of ML-driven insights are intrinsically linked to the quality and consistency of the underlying data. Issues pertaining to sensor calibration, data heterogeneity across sources, and substantial computational resource demands continue to present significant obstacles. Furthermore, the inherent ‘black-box’ nature of certain ML architectures raises valid concerns regarding the interpretability of their outputs, while ethical and privacy implications associated with the collection and analysis of large-scale urban datasets necessitate rigorous attention and the development of responsible data governance frameworks. These limitations underscore the imperative for ongoing refinement of analytical frameworks and the development of more robust and transparent models capable of effectively capturing the intricate complexities of urban environments.

Crowdsourced air temperatures contrast satellite measures of the urban heat island and its mechanisms
The ubiquitous nature of satellite data has led to an explosion of studies on the surface urban heat island (SUHI). Relatively few have simultaneously used air temperature measurements to compare SUHI with the canopy UHI (CUHI), which is more relevant to public health. Using crowdsourced citizen weather stations (>50,000) and satellite data over Europe, we estimate the CUHI and SUHI intensity

in 342 urban clusters during the 2019 heat wave. Satellites produce a sixfold overestimate of UHI relative to station measurements (mean SUHI 1.45°C; CUHI 0.26°C), with SUHI exceeding CUHI in 96% of cities during daytime and in 80% at night. Using empirical evidence, we confirm the control of aerodynamic roughness on UHI intensity, but find evaporative cooling to have a stronger overall impact during this time period. Our results support urban greening as an effective UHI mitigation strategy and caution against relying on satellite data for urban heat risk assessments.

Graph-Based Deep Learning for Short-Term Traffic Flow Forecasting on Urban Arterial Segments

Conventional convolutional neural networks (CNNs), constrained by their reliance on Euclidean data structures, often fall short in capturing the intricate spatial dependencies present in urban traffic systems. To overcome this limitation, this study proposes a graph-based deep learning framework wherein urban arterial road segments are represented as nodes within a graph structure. Their spatial interconnections are then learned through graph convolutional networks (GCNs), which are adept at handling non-Euclidean topologies. By embedding the topological layout of the traffic network into the model architecture, this approach facilitates more precise short-term traffic flow predictions. Comparative experiments reveal that the GCN-based model substantially outperforms traditional methods in modeling inter-segment correlations, particularly in scenarios involving irregular sensor distributions and variable spatial distances. These improvements translate into enhanced forecasting accuracy and model stability, highlighting the potential of graph-based techniques for traffic flow analysis in complex metropolitan settings.

This paper presents a GCN-based framework for short-term traffic flow prediction that effectively addresses the spatial modeling limitations of conventional CNN methods. By abstracting the road network as a graph and leveraging GCNs to learn spatial correlations, the model achieves enhanced accuracy and robustness. Experiments confirm that GCAN captures both topological structure and inter-sensor dependencies more comprehensively than traditional models, successfully handling challenges such as irregular sensor placement, unequal spacing, and disconnected nodes.

The study contributes a practical and scalable method for intelligent traffic management. With continued refinement, the proposed approach holds strong promise for deployment in smart transportation systems and the advancement of future mobility infrastructures.

Geographically weighted random forests for macro-level crash frequency prediction

Machine learning models such as random forests (RF) have been widely applied in the field of road safety. RF is a prominent algorithm, overcoming the limitations of using a single decision tree such as overfitting and instability. However, the traditional RF is a global concept, and thus may fail to capture spatial variability. In macro-level analysis of road safety, the relationship between crash frequency and risk factors can vary spatially. To address this issue, we employ a modified RF algorithm, named geographically weighted random forest (GWRF). Based on the data from London at the level of Middle-super-output-area (MSOA), the predictive

performances of RF and GWRF are compared using mean absolute error (MAE) and root mean square error (RMSE). Moreover, considering MSOAs are geographically connected with each other, several factors related to the discrepancies between adjacent zones are also included in the models. Our results indicate that GWRF outperforms the traditional RF and GWR when an appropriate bandwidth is selected. We further explore the effects of multicollinearity on model performance. The results show that prediction accuracy of GWRF models are not susceptible to the multicollinearity. However, the importance values of those variables with multicollinearity may reduce. Finally, and of equal importance, it is found that the importance of each explanatory variable varies across zones. The density of minor road makes the highest contribution to crash frequency in downtown area, while the crash frequency in peripheral area is more sensitive to the discrepancy of road environment between MSOAs. With such information, road safety interventions can be designed and implemented according to the locally important factors, avoiding thus general guidelines addressed for the entire city.

Positional Encoder Graph Neural Networks for Geographic Data

Graph neural networks (GNNs) provide a powerful and scalable solution for modeling continuous spatial data. However, they often rely on Euclidean distances to construct the input graphs. This assumption can be improbable in many real-world settings, where the spatial structure is more complex and explicitly non-Euclidean (e.g., road networks). Here, we propose PE-GNN, a new framework that incorporates spatial context and correlation explicitly into the models. Building on recent advances in geospatial auxiliary task learning and semantic spatial embeddings, our proposed method (1) learns a context-aware vector encoding of the geographic coordinates and (2) predicts spatial autocorrelation in the data in parallel with the main task. On spatial interpolation and regression tasks, we show the effectiveness of our approach, improving performance over different state-of-the-art GNN approaches. We observe that our approach not only vastly improves over the GNN baselines, but can match Gaussian processes, the most commonly utilized method for spatial interpolation problems.

A Street Graph-Based Morphometric Characterization of Two Large Urban Areas

: Urban microclimate modelling, both numerical and in the laboratory, has strong implications in many relevant health and life-style management issues e.g., in studies for assessment and forecast of air quality (for both outdoor and, as boundary conditions, indoor investigations), for thermometric trend analysis in urban zones, in cultural heritage preservation, etc. Moreover, the study of urban microclimate modelling is largely promoted and encouraged by international institutions for its implication in human health protection. In the present work, we propose and discuss an adaptive street graph-based method aimed at automatically computing the geometrical parameters adopted in atmospheric turbulent flow modelling. This method has been applied to two real cases, the Italian cities of Rome and Cagliari, and its results has been compared with the ones from traditional methods based on regular grids. Results show that the proposed method leads to a more accurate determination of the urban canyon parameters (Canyon Aspect Ratio and Building Aspect Ratio) and morphometric

parameters (Planar Area Index and Frontal Area Index) compared to traditional regular grid-based methods, at least for the tested cases. Further investigations on a larger number of different urban contexts are planned to thoroughly test and validate the proposed algorithm.

The urban air quality modelling, both numerical and in laboratory, needs an urban geometrical parametrization as accurate as possible: in this paper, we have presented and tested a novel street graph-based method for the morphometric and urban canyons parametrization of actual urban areas. This street graph-based method allows to automatically identify the single building or block: in this way, the object of the morphometric analysis is its natural basic component, instead of the arbitrarily grouping and partially decomposing into the square of a grid. This method has been applied to two areas in the center of two Italian cities, Cagliari and Rome, which, being the result of millenniums of urban development, enclose many complex building features. The results, in terms of capability to correctly catch those complexities and of meaningfulness of the parameter values, have been discussed and compared with the ones from traditional regular grids.
Results show a better performance of the proposed method in comparison to the traditional regular grid-based methods. The analysis on the Planar Area Index λp and the Frontal Area Index λf (on every wind direction, averaged over all the wind directions and along the prevailing wind direction), highlights how the regular grid method tends to smooth some of the building features, in particular the ones related to building height, while the street graph-based grid method leads to a better representation of the real, urban texture features of the specific site, so achieving a more realistic representation. Moreover, the regular grid method tends to locate the maxima of λf always along the wind directions related to the grid square diagonals.
Moreover, the adaptive grid leads to more realistic urban shape descriptions for applications that require a downscaling of the parameterization, e.g., not considering the whole city but smaller areas (such as neighborhoods or districts): this is highlighted by the Aspect Ratios analysis. For smaller areas, in fact, the pdfs of the Canyon Aspect Ratio, *ARb,* and of the Building Aspect Ratio, *ARc,* show a different behavior than for larger ones. In particular, pdfs computed over the whole areas depict Cagliari and Rome as two towns with similar building features, whilst the pdfs computed over the single districts are able to highlight the existing differences.
This work therefore provides a simple and rigorous urban area classification tool, very useful both for meso- and micro-scale numerical modeling, and for laboratory scale atmospheric simulations.
Further works are needed and planned: in particular, at this stage, a limitation can be related to the small number of urban areas tested, so it is necessary to test the model on a larger number of datasets, coming from different urban contexts.

Accelerating Large Eddy Simulations of Urban Airflow with Generative Adversarial Networks

This study presents a deep learning framework designed to accelerate Large Eddy Simulation (LES) of airflow in urban environments. The framework leverages physics constrained conditional Generative Adversarial Networks (GANs) trained on instantaneous velocity snapshots from a synthetically generated dataset comprising 130 high-fidelity CFD simulations of simple building configurations. By learning the mapping from early-stage flow fields to their statistically steady-state

counterparts, the framework allows the simulation to bypass the lengthy transient averaging phase and predict the final time-averaged fields directly. Two GAN-based architectures are explored: a conventional convolutional model operating on structured uniform grids (Grid-GAN), and a graph-based model (Graph-GAN) that utilizes Graph Neural Networks (GNNs), specifically Graph Attention Networks (GATs), to process unstructured CFD mesh data while preserving native spatial connectivity. Both approaches are integrated into a fully automated pipeline built exclusively on open-source tools, including OpenFOAM for CFD simulations, FreeCAD and ParaView for preprocessing, and PyTorch for deep learning model development and training. Results demonstrate that the proposed models can significantly reduce LES computational costs while retaining accuracy in predicting turbulent flow characteristics. The Graph-GAN, in particular, shows enhanced adaptability and physical consistency due to its ability to exploit mesh refinements in critical regions. This work lays the foundation for the development of robust, physics-informed surrogate models and supports the growing integration of deep learning with scientific simulations in fluid mechanics.

Adapting machine learning for environmental spatial data - A review

Large-scale modeling of environmental variables is an increasingly complex but necessary task. In this paper, we review the literature on using machine learning to cope with challenges associated with spatial autocorrelation. Our focus was on studies in which researchers predicted environmental variables using a supervised machine learning regression algorithm that accounted for spatial autocorrelation in any part of the modeling pipeline from data exploration to model validation. Methods included explicit spatial covariates, data splitting for training–testing, model calculations, and independent exploratory analysis. Authors accounted for spatial autocorrelation most often in the independent exploratory analysis which had no impact on the predicted values.
We concluded that there seems to be no overall systematic approach on how to account for spatial autocorrelation in machine learning models. In the selected studies, the most appropriate spatial method depended on the specific characteristics of the study. Using explicit spatial covariates and data splitting for training-testing provided more insights into a method's applicability. We summarize these insights and provide considerations for selecting the most appropriate method.

Machine learning (ML) opens up new opportunities to perform environmental modeling. Supervised ML algorithms are versatile tools for various prediction tasks, as they have minimal requirements for the data. They are even more attractive due to their capacity to reveal previously unknown complex relationships in large datasets. However, spatial autocorrelation (SA) needs to be addressed when applying ML to environmental phenomena because it can contribute to biased model training and evaluation scores. In this review, we analyzed 94 papers that incorporated a method that accounted for spatial autocorrelation in supervised ML modeling of environmental phenomena. We grouped these methods into four groups according to their placement in the modeling pipeline: explicit spatial covariates, data splitting for training-testing, model calculations, and independent exploratory analysis. In the majority of studies, authors addressed SA in the independent exploratory analysis, which had no influence on the predicted values.
Currently there seems to be no systematic approach on how to deal with SA in ML.

This was especially evident for methods concerning the model calculations part of the modeling pipeline. More comparative studies on these methods are needed to evaluate the model accuracy and the improvement that can be achieved by incorporating SA. In addition, researchers should study the influence of using one versus multiple methods across the modeling pipeline to explore the potential benefits or drawbacks.
Computation times should also be recorded. Several studies have demonstrated that different ML methods might differ marginally from each other in terms of accuracy, but the computational time differed substantially. Computational efficiency has become relevant because training ML algorithms consumes significant amounts of energy. To reduce the carbon footprint, we should aim for better efficiency in addition to accuracy; that is, we should aim for Green AI/

Attention based spatio-temporal graph convolutional network with focal loss for crash risk evaluation on urban road traffic network based on multi-source risks

The urban road transportation has presented a high probability of crash occurrence, and the aim of the present study is to evaluate the crash risk for urban road networks. However, the irregular structure of urban road networks, the high-dimensional spatio-temporal correlations among multi-source risks (i.e., the contributing risks from traffic flow, meteorological conditions, road design, and so forth), and the issue of data imbalance have brought challenges to this topic. To solve these issues, an Attention based Spatio-Temporal Graph Convolutional Network (ASTGCN) model with focal loss function is used for the first time to evaluate crash risk on an urban road network. This work can be summarized as (1) adopting the spatio-temporal graph convolution structure to capture the spatio-temporal properties and characterize the multi-source risks; (2) utilizing an attention mechanism network to address the critical contributing risks during crash risk evaluation; (3) introducing the focal loss function to improve the model performance impacted by the imbalanced data; and (4) investigating the different contributions of multi-source risks to model performance. The evaluation performance is tested in a real-world urban road traffic network. The raw data consists of 1239 crash records with corresponding datasets of traffic flow characteristics, meteorological conditions, road attributes and the topological structure of the road network. At the same time, three baseline models Artificial Neural Network (ANN), Random Forest (RF), and Deep Spatio-Temporal Graph Convolutional Network (DSTGCN) are compared to the proposed ASTGCN on the same datasets. Overall, the results show that ASTGCN outperforms the baseline models in several evaluation metrics. ASTGCN with focal loss function further improves performance by tackling the issues of dataset imbalance. Additionally, it is also found that the traffic flow risk is most crucial to model performance. The findings of the present study indicate that the proposed model can efficiently evaluate dynamic crash risk for urban road networks, which will benefit the safety management of urban road transportation.

Convolutional neural networks for road surface classification on aerial imagery

Any place the human species inhabits is inevitably modified by them. One of the first features that appear everywhere, in urban areas as well as in the countryside or deep forests, are roads. Further, roads and streets in general reflect their

omnipresent and significant role in our lives through the flow of goods, people, and even culture and information. However, their contribution to the public is highly influenced by their surface. Yet, research on automated road surface classification from remotely sensed data is peculiarly scarce. This work investigates the capacities of chosen convolutional neural networks (fully convolutional network (FCN), U-Net, SegNet, DeepLabv3+) on this task. We find that convolutional neural network (CNN) are capable of distinguishing between compact (asphalt, concrete) and modular (paving stones, tiles) surfaces for both roads and sidewalks on aerial data of spatial resolution of 10 cm. U-Net proved its position as the best-performing model among the tested ones, reaching an overall accuracy of nearly 92%. Furthermore, we explore the influence of adding a near-infrared band to the basic red green blue (RGB) scenes and stress where it should be used and where avoided. Overfitting strategies such as dropout and data augmentation undergo the same examination and clearly show their pros and cons. Convolutional neural networks are also compared to single-pixel based random forests and show indisputable advantage of the context awareness in convolutional neural networks, U-Net reaching almost 25% higher accuracy than random forests. We conclude that convolutional neural networks and U-Net in particular should be considered as suitable approaches for automated semantic segmentation of road surfaces on aerial imagery, while common overfitting strategies should only be used under particular conditions.

Every CNN model ran four times on each of the datasets mentioned in “Training Dataset”; twice in different model settings, twice in different dataset settings. The model setting depended on the dropout rate—one training without dropout layers and one with a dropout rate of 0.5 (50%). The dataset setting depended on data augmentation—one run performed without data augmentation, one with simple data augmentation (using only rotations of 90, 180, and 270 degrees). Altogether, twelve trainings ran for each CNN model, as depicted in Table 1. The band aspects enetring the training are illustrated in Fig. 10. The effect of dropout and data augmentation on the results will be discussed in “Comparison among Methods” and “Comparison among Band Sets”, respectively. Results for different models and configurations are presented and discussed in this section. Overall accuracy values for all CNNs and data settings used in this study are reported in “Comparison among Methods”, where the performance of various CNN architectures and the impact of dropout are discussed, too. “Comparison among Band Sets” details a comparison of the performance of the best architecture across different band sets and the effect of the simple data augmentation.

Graph-Based Deep Learning for Short-Term Traffic Flow Forecasting on Urban Arterial Segments

Conventional convolutional neural networks (CNNs), constrained by their reliance on Euclidean data structures, often fall short in capturing the intricate spatial dependencies present in urban traffic systems. To overcome this limitation, this study proposes a graph-based deep learning framework wherein urban arterial road segments are represented as nodes within a graph structure. Their spatial interconnections are then learned through graph convolutional networks (GCNs), which are adept at handling non-Euclidean topologies. By embedding the topological layout of the traffic network into the model architecture, this approach facilitates more precise short-term traffic flow predictions. Comparative experiments reveal that the GCN-based model substantially outperforms

traditional methods in modeling inter-segment correlations, particularly in scenarios involving irregular sensor distributions and variable spatial distances. These improvements translate into enhanced forecasting accuracy and model stability, highlighting the potential of graph-based techniques for traffic flow analysis in complex metropolitan settings.

This paper presents a GCN-based framework for short-term traffic flow prediction that effectively addresses the spatial modeling limitations of conventional CNN methods. By abstracting the road network as a graph and leveraging GCNs to learn spatial correlations, the model achieves enhanced accuracy and robustness. Experiments confirm that GCAN captures both topological structure and inter-sensor dependencies more comprehensively than traditional models, successfully handling challenges such as irregular sensor placement, unequal spacing, and disconnected nodes.
The study contributes a practical and scalable method for intelligent traffic management. With continued refinement, the proposed approach holds strong promise for deployment in smart transportation systems and the advancement of future mobility infrastructures.

Intra-city scale graph neural networks enhance short-term air temperature forecasting

Air temperature (*T*a) has critical implications for various socioeconomic sectors, yet its dynamics are particularly complex in urban areas due to heterogeneous built environments, landscapes, and diverse anthropogenic activities. Physics-based models struggle with intra-city *T*a forecasts due to inadequate urban representation and limited spatial resolution. While weather observation networks offer promising alternatives for direct modeling with local *T*a time-series, an effective framework to leverage these intra-city discrete sensor data remains lacking. Here, we demonstrate that graph neural networks (GNNs) can harness observation network information to refine *T*a prediction at individual locations and elucidate underlying mechanisms. Our novel Mix-n-Scale framework with GNNs achieves over 12 % improvement in short-term *T*a forecasts compared to conventional local time-series approaches. Further model evaluation disentangles performance variations with local *T*a variability in diverse spatiotemporal contexts, indicating distinct patterns of intra-city heterogeneity across seasonal and diurnal scales. Our findings establish graph-based approaches for leveraging proliferating urban sensor data and advancing understanding of *T*a spatiotemporal dynamics in complex urban environments.

Machine learning methods and approaches for Urban Heat Island (UHI) assessment: A comprehensive review

The Urban Heat Island (UHI) effect is a phenomenon in which urban areas experience significantly higher temperatures than their surrounding regions due to human activities and the built environment. This poses challenges to urban systems, living conditions, and exacerbate air pollution in urban areas. Over the years, the use of Machine Learning (ML) to study the UHI effect has increased, driven by advances in computational power, high predictive accuracy, and the growing availability of large-scale, environmental datasets. This study aims to comprehensively review the application and role of ML in addressing the UHI effect.

The key findings of this review show that the application of ML can be categorized into: land surface temperature (LST) retrieval and prediction, land use land cover (LULC) classification and change detection, UHI intensity and pattern modelling, and mitigation strategy assessment. The most common application is the derivation and prediction of LST in conjunction with physical and climatic variables such as building energy consumption, air temperature, human thermal comfort, and wind movement. Our findings further show that ML is effectively utilized in multi-temporal investigations and predictive modelling, assessing energy demand surges, and validating the efficacy of cooling interventions. We also outline the key challenges faced in implementing ML in this realm, including aerial data accessibility, data pre-processing complexities, modelling sophistication or UHI, and lack of ground truth data for validation. This review sets the stage for future use of ML, including emerging Artificial Intelligence (AI) - driven methods that could change how we understand and mitigate UHI effects.

ML has demonstrated remarkable potential to advance UHI assessment, particularly by enhancing the predictive accuracy for LST and uncovering complex nonlinear relationships intrinsic to urban thermal environments. Recent meta-analyses underscore that ML algorithms, such as hybrid CNN-LSTM, substantially outperform traditional approaches, delivering reductions in RMSE by 23–41% and maintaining tight error margins while dramatically lowering computational loads. However, the widespread adoption of ML in UHI research remains impeded by several persistent limitations. Notably, data heterogeneity - stemming from the integration of disparate sensor networks and variable spatiotemporal resolutions - presents significant challenges, especially when fusing high-resolution satellite imagery with granular IoT sensor data. The opacity of contemporary DL models poses another major obstacle, as limited interpretability restricts causal inference regarding the influence of urban morphology and green infrastructure on UHI dynamics. Further, validation bottlenecks prevail, given that only a minority of studies employ robust UAV-based thermal validation, with most relying instead on indirect satellite data cross-validation.

Fine-scale modeling of the urban heat island: A comparison of multiple linear regression and random forest approaches

Characterizing the spatiotemporal variability of the Urban Heat Island (UHI) and its drivers is a key step in leveraging thermal comfort to create not only healthier cities, but also to enhance urban resilience to climate change. In this study, we developed specific daytime and nighttime multiple linear regression (MLR) and random forest (RF) models to analyze and predict the spatiotemporal evolution of the Urban Heat Island intensity (UHII), using the air temperature (Tair) as the response variable. We profited from the wealth of *in situ* Tair data and a comprehensive pool of predictors variables — including land cover, population, traffic, urban geometry, weather data and atmospheric vertical indices. Cluster analysis divided the study period into three main groups, each dominated by a combination of weather systems that, in turn, influenced the onset and strength of the UHII. Anticyclonic circulations favored the emergence of the largest UHII (hourly mean of 5.06 °C), while cyclonic circulations dampened its development. The MLR models were only able to explain a modest percentage of variance (64 and 34% for daytime and nighttime, respectively), which we interpret as part of their inability to capture key factors controlling Tair. The RF models, on the other

hand, performed considerably better, with explanatory power over 96% of the variance for daytime and nighttime conditions, capturing and mapping the fine-scale Tair spatiotemporal variability in both periods and under each cluster condition. The feature importance analysis showed that the meteorological variables and the land cover were the main predictors of the Tair. Urban planners could benefit from these results, using the high-performing RF models as a robust framework for forecasting and mitigating the effects of the UHI.

Method to predict urban heat island hot spots at microscale level based on convolutional neural network land cover

The ongoing shifts in climate change suggest an increase in urban temperatures during hot periods, emphasising the need for supported public policy to be more efficient in reducing Urban Heat Island effects. This paper presents a CNN-based method to classify land cover and simulate micro-scale temperature variations for identifying UHI hot spots. The five primary urban heat sources – buildings, grass, trees, roads, and water – are analysed for their impact on local temperatures. Using a dataset of 775,000 tiles (60 × 60 pixels), dedicated CNN models are trained for each class. Model validation achieved over 97% accuracy for all classes except roads, which reached 89.1%. Simplified microscale models simulate temperature variations, producing a map that visualises UHI hot spots across urban areas. The results highlight the tool's effectiveness and practicality in identifying high-temperature zones, offering valuable support for UHI mitigation and targeted policy efforts.

Microclimate prediction for sandy photovoltaic power plants using a spatio-temporal graph convolutional network with environmental covariates

The deployment of photovoltaic (PV) power plants in sandy regions alters the local microclimate. Accurate microclimate prediction is essential for optimizing short-term operations and maintenance plans. Compared with traditional statistical and numerical models, neural networks are suited to capture the complex nonlinear dependencies in sandy-region PV environments. Graph neural networks (GNNs) extend neural networks to process data defined on graph domains, enabling them to confront high-dimensional feature redundancy, attenuated temporal dependencies, and pronounced spatial heterogeneity. Accordingly, we propose a spatio-temporal graph convolutional network with environmental covariates (EC-STGCN) for station-level microclimate prediction at meteorological stations. Experimental results demonstrate that EC-STGCN achieves accurate high-resolution spatio-temporal predictions of temperature, relative humidity (RH), and particulate matter (PM_{10}) and shows improved predictive performance relative to conventional GNNs, while covariate-level analysis shows that individual environmental covariates contribute heterogeneously to these predictions.

Predicting high-resolution air quality using machine learning: Integration of large eddy simulation and urban morphology data

Accurately predicting air pollutants, especially in urban areas with well-defined spatial structures, is crucial. Over the past decade, machine learning techniques have been widely used to forecast urban air quality. However, traditional machine

learning approaches have limitations in accuracy and interpretability for predicting pollutants. In this study, we propose a convolutional neural network (CNN) model to predict the spatial distribution of CO concentration in Nanjing urban area at 10 m resolution. Our model incorporates various factors as input, such as building height, topography, emissions, and is trained against the outputs simulated by the parallelized large-eddy simulation model (PALM). The PALM model has 48 different scenarios that varied in emissions, wind speeds, and wind directions. The results display a strong consistency between the two models. Furthermore, we evaluate the performance of our model using a 10-fold cross-validation and out-of-sample cross-validation approach. This yields a robust correlation (with both $R2 > 0.8$) and a low RMSE between the CO predicted by the PALM and CNN models, which demonstrates the generalization capability of our CNN model. The CNN can extract crucial features from the resulted weight contribution map. This map indicates that the CO concentration at a location is more influenced by nearby buildings and emissions than distant ones. The interpretable patterns uncovered by our model are related to neighborhood effects, wind speeds, directions, and the impact of orientation on urban CO distribution. The model also shows high prediction accuracy ($R > 0.8$) when applied to another city. Overall, the integration of our CNN framework with the PALM model enhances the accuracy of air quality predictions, while enabling a fluid dynamic laws interpretation, providing effective tools for air quality management.

Reverse Thinking: A New Method from the Graph Perspective for Evaluating and Mitigating Regional Surface Heat Islands

Abstract

Accurately locating key nodes and corridors of an urban heat island (UHI) is the basis for effectively mitigating a regional surface UHI. However, we still lack appropriate methods to describe it, especially considering the interaction between UHIs and the role of connectivity (network). Specifically, previous studies paid much attention to the raster and vector perspective—based on standard landscape configuration metrics that only provide an overall statistic over the entire study area without further indicating locations where different types of pattern and fragmentation occur. Therefore, by reverse thinking, here we attempt to propose a new method from the graph perspective which integrates morphological spatial pattern analysis (MSPA)—which is used to characterize binary patterns with emphasis on connections between their parts as measured at varying analysis scales, and habitat availability indices to evaluate and mitigate regional surface UHI. We selected the Pearl River Delta Metropolitan Region (PRDR), one of the most rapidly urbanized regions in the world as the case study (1995–2015). The results of the case study showed: (1) the core (UHI) type accounts for the vast majority of the MSPA model, with the relative land surface temperature (LST) rises, the proportion of the core type will increase, and it could influence the edge (UHI) type significantly; (2) the branch, bridge, and islet (UHI) types have similar results to the lower temperature ($4 < \text{Relative LST} \leq 6$) area and account for the majority, indicating that these types are more susceptible to their surrounding environment; (3) the importance and extreme importance area (node) from 1995 to 2015 have increased significantly and mainly distributed in the urbanized areas, which means cooling measures need to be implemented in these areas in order of priority. Shifting the research logic of UHI evaluation and mitigation from “patch” to “network”, we hold the point that the method (reverse thinking) has significant theoretical and practical implications for mitigating regional UHI and urban climate-resilience.

Spatial heterogeneity effect of built environment on traffic safety using geographically weighted atrous convolutions neural network

The built environment exerts a significant influence on the frequency and severity of traffic accidents. Spatially uniform assumptions on the impacts of built environment factors commonly employed in existing research may lead to inconsistent and contradictory findings. While some studies have investigated spatial heterogeneity using geographically weighted regression models (GWR), these approaches frequently neglect critical aspects including the road network distance between built environment features and the non-linear decay of influence relationships. To address these methodological limitations, this study develops a geographically weighted atrous convolutional neural network regression model (GACNNWR) to more accurately capture the spatial heterogeneity in the impact of built environment factors on traffic safety. Based on empirical data of traffic accidents and built environment from Jinan City, our results demonstrate that the GACNNWR model outperforms traditional analytical methods such as GWR model. Intersection density and bus stop density are identified as having a more substantial impact on traffic accidents compared to population density, land use mix, and destination accessibility. Additionally, population density is shown to exert a bidirectional influence on traffic accidents, while the spatial variability in the effects of land use mix is relatively pronounced. These findings provide important implications for the design of context-sensitive built environments and the formulation of localized traffic safety management strategies aimed at mitigating crash risks.

Spatial prediction of environmental processes using random forests: How best to account for spatial dependence?

Geostatistical spatial prediction for environmental processes is typically undertaken using Gaussian process models via Kriging, while machine learning (ML) algorithms are state-of-the-art for non-spatial prediction. An exciting recent fusion of these ideas imbibes traditional ML algorithms with the capacity to deal with spatial autocorrelation, leading to improved predictive performance. A range of approaches have been proposed, including fusion with Gaussian processes, observation-driven correlation structures, spatial basis functions and local geographical fitting. However, there has been no numerical comparison of their relative predictive performances, which is needed to advise environmental scientists on the optimal approach to use. This paper fills this knowledge gap, and focuses on random forests as the ML algorithm because they are more computationally and conceptually straightforward to implement than deep learning algorithms. The results from two studies are presented, the first being a controlled simulation experiment investigating whether any single approach is consistently superior across different spatial autocorrelation types. The second study focuses on the prediction of air pollution concentrations within a tuberculosis prevalence study in Blantyre, Malawi. The results show that whilst no single approach is universally superior, utilising spatial basis functions appears to perform consistently well across both the simulation and real data studies.

Spatially Aware Deep Learning for Microclimate Prediction from High-

Resolution Geospatial Imagery

Microclimate models are essential for linking climate to ecological processes, yet most physically based frameworks estimate temperature independently for each spatial unit and rely on simplified representations of lateral heat exchange. As a result, the spatial scales over which surrounding environmental conditions influence local microclimates remain poorly quantified. Here, we show how remote sensing can help quantify the contribution of spatial context to microclimate temperature predictions. Building on convolutional neural network principles, we designed a task-specific deep neural network and trained a series of models in which the spatial extent of input data was systematically varied. Drone-derived spatial layers and meteorological data were used to predict ground temperature at a focal location, allowing direct assessment of how prediction accuracy changes with increasing spatial context. Our results show that incorporating spatially adjacent information substantially improves prediction accuracy, with diminishing returns beyond spatial extents of approximately 5-7 m. This characteristic scale indicates that ground temperatures are influenced not only by local surface properties, but also by horizontal heat transfer and radiative interactions operating across neighboring microhabitats. The magnitude of spatial effects varied systematically with time of day, microhabitat type, and local environmental characteristics, highlighting context-dependent spatial coupling in microclimate formation. By treating deep learning as a diagnostic tool rather than solely a predictive one, our approach provides a general and transferable method for quantifying spatial dependencies in microclimate models and informing the development of hybrid mechanistic-data-driven approaches that explicitly account for spatial interactions while retaining physical interpretability.

The study developed a deep neural network framework to quantify the importance of spatial context in microclimate temperature predictions. By systematically varying the spatial extent of model inputs, we showed that incorporating nearby environmental information substantially improves predictive accuracy, with gains saturating at spatial scales of several meters. The strength of these effects varied across times of day, microhabitats, and spatial characteristics, indicating that 25 horizontal heat transfer and spatial heterogeneity play a context-dependent role in shaping ground temperatures. Rather than serving solely as a predictive tool, deep learning allowed us to identify when and where simplifying assumptions in traditional microclimate models are most likely to break down. Together, our findings highlight the value of integrating high-resolution spatial data with modeling approaches that explicitly account for spatial interactions, and they provide a foundation for developing hybrid mechanistic–machine learning microclimate models that are both physically grounded and ecologically relevant.

Spatiotemporal Graph Learning on Urban Environments

Urban environments are shaped by intricate interactions among water, soil, air, and infrastructure, where traditional models often fail to capture nonlinear, non-Euclidean dynamics. Spatiotemporal graph learning (STGL) has emerged as a powerful framework to represent such complexity, enabling accurate forecasting and real-time decision support from urban districts to national and even global scales. This review provides the first comprehensive synthesis of STGL tailored to urban environments. We summarize advances in graph construction, spatial and temporal modeling, and fusion strategies, and examine applications across urban

water systems, soil and agriculture, air quality, and urban risk. Landmark case studies, including Microsoft's Aurora, NVIDIA's Earth-2, and Google's GraphCast/GenCast, demonstrate STGL's potential as a foundation model for environmental intelligence. We conclude by identifying key limitations and outlining future directions, emphasizing federated learning, machine unlearning, and meta-learning to enhance next-generation STGL frameworks that ultimately support resilient and adaptive urban environments.

Street-level temperature estimation using Graph Neural Networks: Performance, feature embedding and interpretability

Estimating street-level air temperature is a challenging task due to the highly heterogeneous urban surfaces, canyon-like street morphology, and the diverse physical processes in the built environment. Though pioneering studies have embarked on investigations via data-driven approaches, many questions remain to be answered. In this study, we leveraged an innovative framework and redefined the street-level temperature estimation problem using Graph Neural Networks (GNN) with spatial embedding techniques. The results showed that GNN models are more capable and consistent of estimating street-level temperature among tested locations, benefiting from its unique strength in handling extensive data over unstructured graph topology. In addition, we conducted an in-depth analysis of feature importance to enhance the model interpretability. Among the urban features analyzed in this study, the time-variant canopy density and meter-level land use data emerge as crucial factors. Our findings highlight GNN's high potential in capturing the complex dynamics between urban elements and their impacts on microclimate, thus offering valuable insights for comprehensive urban data collection and urban climate modeling in general. Collectively, this study also contributes to urban planning and policy by providing avenues to enhance city resilience against climate change, thereby advancing the agenda for environmental stewardship and urban sustainability.

Super-Resolution of Three-Dimensional Temperature and Velocity for Building-Resolving Urban Micrometeorology Using Physics-Guided Convolutional Neural Networks with Image Inpainting Techniques

Atmospheric simulations for urban cities can be computationally intensive because of the need for high spatial resolution, such as a few meters, to accurately represent buildings and streets. Deep learning has recently gained attention across various physical sciences for its potential to reduce computational cost. Super-resolution is one such technique that enhances the resolution of data. This paper proposes a convolutional neural network (CNN) that super-resolves instantaneous snapshots of three-dimensional air temperature and wind velocity fields for urban micrometeorology. This super-resolution process requires not only an increase in spatial resolution but also the restoration of missing data caused by the difference in the building shapes that depend on the resolution. The proposed CNN incorporates gated convolution, which is an image inpainting technique that infers missing pixels. The CNN performance has been verified via supervised learning utilizing building-resolving micrometeorological simulations around Tokyo Station in Japan. The CNN successfully reconstructed the temperature and velocity fields around the high-resolution buildings, despite the missing data at lower altitudes due to the coarseness of the low-resolution buildings. This result

implies that near-surface flows can be inferred from flows above buildings. This hypothesis was assessed via numerical experiments where all input values below a certain height were made missing. This research suggests the possibility that building-resolving micrometeorological simulations become more practical for urban cities with the aid of neural networks that enhance computational efficiency.

The Effect of Urban Geometry on Microclimate

The aim of this paper is to investigate the effect of urban geometry on microclimate through a case study of the Greater London area. GIS data are used to describe the urban characteristics and calculate both spatial average and heterogeneity variables. Computational Fluid Dynamics (CFD) simulations are performed in areas of London in order to obtain their microclimatic conditions. The CFD simulations are performed using the standard kε turbulence model and Reynolds-Averaged Navier Stokes (RANS) equations. Next, correlation analyses using the Spearman's rank correlation coefficient are carried out to determine the impact of the urban variables on microclimate. The case study demonstrates that average urban density and building geometry have a substantial effect on microclimate. Furthermore, variation in building geometry and canyon ratio also have a significant impact. It is thus concluded that both average spatial variables and spatial heterogeneity should be included in urban climate studies.

Keywords: urban, microclimate, CFD, London

Uncertainty-Aware Graph Neural Reconstruction of Urban Temperature Fields from Sparse Sensors under Deployment Constraints

deployments are limited by sensor budgets and spacing constraints. This study proposes an uncertainty-aware graph neural network (GNN) framework for reconstructing daily maximum temperature fields from sparse sensors while supporting distance-constrained sensor placement and probabilistic exceedance mapping. The model predicts both the temperature field and a spatially varying predictive uncertainty field using a graph-attention-based mean-residual architecture trained with a Gaussian negative log-likelihood. Sensor placement is addressed using a Proper Orthogonal Decomposition with QR factorization (POD-QR) strategy with a 4 km minimum inter-sensor distance constraint and is compared with random feasible placement and farthest-point sampling. The framework is evaluated over a Montreal-area polygon using Daymet v4.1 daily temperature data (1 km resolution) under a strict temporal hold-out protocol (training: 2020-2023; testing: 2024). Across sensor budgets (10-40 sensors), the proposed GNN consistently outperforms inverse distance weighting and ordinary kriging in RMSE and MAE on unobserved nodes. Sensor-placement effects are most pronounced at low budgets and diminish at higher budgets, with a practical saturation regime emerging around 30 sensors under the imposed spacing constraint. Probabilistic evaluation further shows improved uncertainty calibration with increasing sensor density and a better sharpness-calibration trade-off than kriging. These results support the proposed framework as an effective tool for uncertainty-aware temperature field reconstruction and decision-oriented heat-risk mapping.

UrbanGraph: Physics-Informed Spatio-Temporal Dynamic Heterogeneous Graphs for Urban Microclimate Prediction

With rapid urbanization, predicting urban microclimates has become critical, as it affects building energy demand and public health risks. However, existing generative and homogeneous graph approaches fall short in capturing physical consistency, spatial dependencies, and temporal variability. To address this, we introduce UrbanGraph, a framework founded on a novel structure-based inductive bias. Unlike implicit graph learning, UrbanGraph transforms physical first principles into a dynamic causal topology, explicitly encoding time-varying causalities (e.g., shading and convection) directly into the graph structure to ensure physical consistency and data efficiency. Results show that UrbanGraph achieves state-of-the-art performance across all baselines. Specifically, the use of explicit causal pruning significantly reduces the model's floating-point operations (FLOPs) by 73.8% and increases training speed by 21% compared to implicit graphs. Our contribution includes the first high-resolution benchmark for spatio-temporal microclimate modeling, and a generalizable explicit topological encoding paradigm applicable to urban spatio-temporal dynamics governed by known physical equations.

SOLUTION

A Lightweight Spatio-Temporal Neural Network With Sampling-Based Time Series Decomposition for Traffic Forecasting

The escalating challenges in urban traffic highlight the importance of accurate traffic forecasting for enhancing the Intelligent Transportation Systems (ITS), given its pivotal role in optimizing traffic management strategies. Due to the impressive performance of Graph Neural Networks in spatio-temporal data analysis, Spatio-Temporal Graph Neural Networks (STGNNs) have emerged as a mainstream approach for traffic forecasting. However, many recent STGNN-based works often utilize redundant and intricate architectures, leading to increased computational costs with only minor improvements in performance. To solve this issue, we propose a concise but effective Lightweight Spatio-Temporal Neural Network architecture, LSTNN. Specifically, we design two sampling strategies, downsampling and patched sampling, to decompose the time series and separately fit the seasonal component and trend components. Based on sampling strategies, we introduce a spatio-temporal localization module based on the raw traffic data and learnable spatio-temporal indexing. In particular, this module models spatio-temporal correlations by appending the spatio-temporal indexing to the traffic sequences. We visualized the spatio-temporal indexing, confirming the effectiveness of the spatio-temporal localization module design. Extensive experiments are conducted on four widely utilized real-world traffic datasets, showing the significant improvement of LSTNN compared to baselines, including several state-of-the-art models. Furthermore, we conducted extensive experiments on four long-term time series datasets to validate the effectiveness of LSTNN in the Long-term Time Series Forecasting (LTSF) task. Our findings may inspire and encourage the design of more concise neural prediction architectures for real-world applications.

A Lightweight Spatiotemporal Graph Framework Leveraging Clustered Monitoring Networks and Copula-Based Pollutant Dependency for PM2.5 Forecasting

Air pollution threatens human health and ecosystems, making timely forecasting essential. The spatiotemporal dynamics of pollutants, shaped by various factors, challenge traditional methods. Therefore, spatiotemporal graph-based deep learning has gained attention for its ability to capture spatial and temporal dependencies within monitoring networks. However, many existing models, despite their high predictive accuracy, face computational complexity and scalability challenges. This study introduces clustered and lightweight spatio-temporal graph convolutional network with gated recurrent unit (ClusLite-STGCN-GRU), a hybrid model that integrates spatial clustering based on pollutant time series for graph construction, Copula-based dependency analysis for selecting relevant pollutants to predict PM2.5, and graph convolution combined with gated recurrent units to extract spatiotemporal features. Unlike conventional approaches that require learning or dynamically updating adjacency matrices, ClusLite-STGCN-GRU employs a fixed, simple cluster-based structure. Experimental results on Tehran air quality data demonstrate that the proposed model not only achieves competitive predictive performance compared to more complex models, but also significantly reduces computational cost—by up to 66% in training time, 83% in memory usage, and 84% in number of floating-point operations—making it suitable for real-time applications and offering a practical balance between accuracy, interpretability, and efficiency.

A Graph Neural Network Approach for Localized and High-Resolution Temperature Forecasting

Heatwaves are intensifying worldwide and are among the deadliest weather disasters. The burden falls disproportionately on marginalized populations and the Global South, where under-resourced health systems, exposure to urban heat islands, and the lack of adaptive infrastructure amplify risks. Yet current numerical weather prediction models often fail to capture micro-scale extremes, leaving the most vulnerable excluded from timely early warnings. We present a Graph Neural Network framework for localized, high-resolution temperature forecasting. By leveraging spatial learning and efficient computation, our approach generates forecasts at multiple horizons, up to 48 hours. For Southwestern Ontario, Canada, the model captures temperature patterns with a mean MAE of 1.93°C across 1-48h forecasts and MAE@48h of 2.93°C, evaluated using 24h input windows on the largest region. While demonstrated here in a data-rich context, this work lays the foundation for transfer learning approaches that could enable localized, equitable forecasts in data-limited regions of the Global South.

A Systematic Literature Review of Spatio-Temporal Graph Neural Network Models for Time Series Forecasting and Classification

In recent years, spatio-temporal graph neural networks (GNNs) have attracted considerable interest in the field of time series analysis, due to their ability to capture, at once, dependencies among variables and across time points. The objective of this systematic literature review is hence to provide a comprehensive overview of the various modeling approaches and application domains of GNNs for time series classification and forecasting. A database search was conducted, and 366 papers were selected for a detailed examination of the current state-of-the-art in the field. This examination is intended to offer to the reader a comprehensive

review of proposed models, links to related source code, available datasets, benchmark models, and fitting results. All this information is hoped to assist researchers in their studies. To the best of our knowledge, this is the first and broadest systematic literature review presenting a detailed comparison of results from current spatio-temporal GNN models applied to different domains. In its final part, this review discusses current limitations and challenges in the application of spatio-temporal GNNs, such as comparability, reproducibility, explainability, poor information capacity, and scalability. This paper is complemented by a GitHub repository at https://github.com/FlaGer99/SLR-Spatio-Temporal-GNN.git providing additional interactive tools to further explore the presented findings.

A unified deep learning framework integrating OpenStreetMap for multi-domain urban planning tasks

Urban planning increasingly requires multi-source geospatial intelligence to address the complexity of modern cities. This study introduces a unified deep learning framework integrating OpenStreetMap (OSM) data with multispectral satellite imagery and demographic-environmental datasets. The framework employs task-specific architectures, including Convolutional Neural Networks (CNNs) for land-use classification, U-Net segmentation for building footprint extraction, Long Short-Term Memory (LSTM) networks for traffic flow prediction, and a hybrid CNN-RNN model for air-quality forecasting. The land-use model achieved 91.6% accuracy, the U-Net building footprint extractor reached 94.0% accuracy, the LSTM traffic model obtained an RMSE of 3.6 vehicles per hour, and the hybrid air-quality model achieved an RMSE of 2.3 µg/m3. These metrics reflect performance on OSM-aligned datasets specifically curated for Krasnodar. The novelty of this work lies in establishing a multi-task, multi-source analytical workflow that unifies OSM, satellite, and temporal environmental data within a coherent deep learning system. The resulting framework provides urban planners with a scalable and low-cost approach for generating high-resolution urban intelligence to support sustainable city management.

A variable-resolution unstructured spatio-temporal graph neural network: an application to very short-range weather forecasting in Guangdong, China

The increase in extreme weather events has raised the demand for very short-range forecasts in Guangdong, the most populous province in China. Although global AI-based weather forecasting models provide strong medium-range guidance, they remain suboptimal for regional very short-range predictions. To address this, we developed a spatio-temporal graph neural network (STGNN) for Guangdong, featuring an unstructured, variable-resolution graph refined over the Pearl River Delta. The core of the network is 1-D dilated causal temporal convolutions and spatial convolutions on combined static and dynamic graphs. A spatial mean constraint and Laplacian-residual regularization were incorporated into the loss function to enhance physical consistency. Trained on historical reanalysis dataset, the model generates hourly forecasts for multiple near-surface atmospheric variables for the next six hours. Verification against an independent test-set and observations, using multiple metrics, demonstrates high predictive skill and strong spatio-temporal coherence for 2 m temperature and sea-level pressure. For 10 m winds and 2 m humidity, model errors grow larger with lead time. Case studies of an extreme heatwave event in Guangzhou and the post-

landfall evolution of Super Typhoon Saola effectively capture the evolution characteristics and spatial patterns, while underestimates the peak temperatures, maximum winds and minimum sea-level pressures. These results support the use of variable-resolution STGNN as a practical approach for very short-range regional forecasting in Guangdong, China.

OSM+: Billion-Level OpenStreetMap Dataset for City-wide Experiments

Road network data provides rich information about cities, but processing worldwide OpenStreetMap (OSM) data is computationally intensive, and the resulting graphs are often difficult to unify for benchmarking downstream tasks. Existing graph learning benchmarks fail to capture the billion-scale and unique topological properties of real-world road networks, leaving model scalability underexplored. To close this gap, we process OSM data with distributed cloud computing using 5,000 cores and release \textbf{OSM+}, a structured worldwide 1-billion-vertex road network graph dataset designed for high accessibility and usability. OSM+ is open source and globally downloadable, providing an open-box graph structure and an easy spatial query interface; the evaluated release is a fixed snapshot for reproducibility, with a versioned update plan for future releases. We demonstrate the utility of OSM+ through three illustrative use cases: city boundary detection, traffic prediction, and traffic policy control. For traffic prediction, we construct a new 31-city benchmark by processing traffic data and combining it with OSM+, enabling broader spatial coverage and more comprehensive evaluation than commonly used datasets, while scaling from hundreds of road network intersections to thousands. For traffic policy control, we release a new six-city dataset at a much larger scale, introducing challenges for thousand-scale multi-agent coordination. We also provide data processing tools for integrating multimodal spatial-temporal data with OSM+ for geospatial foundation model training, thereby expediting the discovery of compelling scientific insights.

OSMGraphCLIP: Learning Global Location Representations from OpenStreetMap Graphs

We present OSMGraphCLIP, a CLIP-style geospatial representation model that learns global location embeddings from freely available OpenStreetMap (OSM) data. OSMGraphCLIP represents geographic environments as heterogeneous graphs of typed OSM features, preserving the topological and semantic relationships among roads, buildings, land-use regions, and points of interest. A multi-scale graph encoder captures both fine-grained local structure and broader landscape composition, and supervises a spherical-harmonics location encoder through a contrastive alignment objective. We evaluate OSMGraphCLIP across a diverse suite of downstream geospatial regression and classification tasks spanning climate, ecology, socioeconomic indicators, public health, land cover, biodiversity, and wildfire forecasting, and show that structured OSM data alone supports strong global location representations across domains. OSMGraphCLIP matches or exceeds satellite-based baselines on the majority of benchmarks, with the most pronounced advantage on socioeconomic and public-health tasks, where OSM's explicit semantic annotation of the built environment encodes patterns of human activity that satellite pixels can only capture indirectly. On ecological and environmental tasks, the model remains closely competitive with imagery-based methods despite using no Earth observation data. Qualitative analysis confirms

that the learned embeddings organize geographic space coherently, recovering biome boundaries, urban gradients, and tropical--temperate distinctions from map topology alone.

Scalable spatio-temporal autoregressive modeling by designing equivalent graph convolutional neural networks

Spatial and spatio-temporal autoregressive models (SAR and STAR) explicitly conceptualize outcomes at one location as a function of outcomes at nearby locations and characterize substantial spatial dependencies. However, their application is constrained by huge computational burdens associated with model estimation, rendering them infeasible for large-scale real-world problems. To address this challenge, we proposed a novel architecture that extends the principles of graph convolutional neural networks to integrate spatial autoregression in neural networks (SARNNs). By establishing the mathematical and algorithmic equivalence between SARNNs and SAR, our approach ensures unbiased parameter estimations while reducing computational complexity to O(n) and makes SAR models scalable. Then, we developed STARNNs, a spatio-temporal extension of SARNNs, by properly stacking SARNNs to form an architecture equivalent to STAR models. The strategy enables efficient and tractable solutions for large-scale complex STAR models. Monte Carlo simulation experiments confirm the equivalence between SAR (STAR) and SARNN (STARNN) modeling and emphasize the critical role of a specifically designed loss function that explicitly incorporates spatial autocorrelation. With STARNNs, we implemented a high-resolution global-scale STAR model to assess economic losses induced by climate change. The results underscore the importance of considering spatial propagation effects to accurately quantify the economic consequences of climate change.

Spatial-Temporal Synchronous Graph Convolutional Networks: A New Framework for Spatial-Temporal Network Data Forecasting

Spatial-temporal network data forecasting is of great importance in a huge amount of applications for traffic management and urban planning. However, the underlying complex spatial-temporal correlations and heterogeneities make this problem challenging. Existing methods usually use separate components to capture spatial and temporal correlations and ignore the heterogeneities in spatial-temporal data. In this paper, we propose a novel model, named Spatial-Temporal Synchronous Graph Convolutional Networks (STSGCN), for spatial-temporal network data forecasting. The model is able to effectively capture the complex localized spatial-temporal correlations through an elaborately designed spatial-temporal synchronous modeling mechanism. Meanwhile, multiple modules for different time periods are designed in the model to effectively capture the heterogeneities in localized spatial-temporal graphs. Extensive experiments are conducted on four real-world datasets, which demonstrates that our method achieves the state-of-the-art performance and consistently outperforms other baselines.

Spatio-Temporal Graph Neural Networks for Predictive Learning in Urban Computing: A Survey

With recent advances in sensing technologies, a myriad of spatio-temporal data has been generated and recorded in smart cities. Forecasting the evolution patterns of spatio-temporal data is an important yet demanding aspect of urban computing, which can enhance intelligent management decisions in various fields, including transportation, environment, climate, public safety, healthcare, and others. Traditional statistical and deep learning methods struggle to capture complex correlations in urban spatio-temporal data. To this end, Spatio-Temporal Graph Neural Networks (STGNN) have been proposed, achieving great promise in recent years. STGNNs enable the extraction of complex spatio-temporal dependencies by integrating graph neural networks (GNNs) and various temporal learning methods. In this manuscript, we provide a comprehensive survey on recent progress on STGNN technologies for predictive learning in urban computing. Firstly, we provide a brief introduction to the construction methods of spatio-temporal graph data and the prevalent deep-learning architectures used in STGNNs. We then sort out the primary application domains and specific predictive learning tasks based on existing literature. Afterward, we scrutinize the design of STGNNs and their combination with some advanced technologies in recent years. Finally, we conclude the limitations of existing research and suggest potential directions for future work.

Spatiotemporal Graph Learning on Urban Environments

Urban environments are shaped by intricate interactions among water, soil, air, and infrastructure, where traditional models often fail to capture nonlinear, non-Euclidean dynamics. Spatiotemporal graph learning (STGL) has emerged as a powerful framework to represent such complexity, enabling accurate forecasting and real-time decision support from urban districts to national and even global scales. This review provides the first comprehensive synthesis of STGL tailored to urban environments. We summarize advances in graph construction, spatial and temporal modeling, and fusion strategies, and examine applications across urban water systems, soil and agriculture, air quality, and urban risk. Landmark case studies, including Microsoft's Aurora, NVIDIA's Earth-2, and Google's GraphCast/GenCast, demonstrate STGL's potential as a foundation model for environmental intelligence. We conclude by identifying key limitations and outlining future directions, emphasizing federated learning, machine unlearning, and meta-learning to enhance next-generation STGL frameworks that ultimately support resilient and adaptive urban environments.

High-Resolution Forecasting of Road Surface Temperature Using Graph Neural Networks

Effective winter road maintenance (WRM) is critical for the safety and mobility of road users in cold regions. High spatial resolution predictions of road surface conditions greatly improve WRM efficiency. Previous studies have explored spatial modeling to achieve this; however, traditional statistical models, such as Kriging, face limitations in capturing complex spatial dependency. In this study, we utilized graph neural networks (GNNs) for spatial modeling to predict road surface temperature (RST), which an essential road weather parameter, at high spatial resolution. Furthermore, by integrating this approach with time-series prediction model at road weather observatories and weather mesh forecast data, we propose

a framework that enables high spatial-resolution RST predictions over future horizons. Experiments using real-world data from Sapporo, Japan demonstrated that the proposed method could predict RST at approximately \mathbf1 0 0-meter intervals along road networks with an root mean square error below 2° C for future horizons up to \mathbf2 4 h. In conclusion, the study findings represent a significant improvement in accuracy over current operational methods, offering valuable support for decision-making in WRM planning to enhance road safety and mobility.

An Open-Source Deep Learning Framework for Scalable Urban Heat Island Detection Using Geospatial Data

Urban Heat Islands (UHIs), where urban areas exhibit elevated temperatures relative to their rural surroundings, pose growing challenges in the context of climate change, particularly for densely built, vegetation-scarce cities. Traditional methods for UHI detection, often based on empirical indices or statistical regressions, lack spatial resolution, scalability, and adaptability across diverse urban environments. This study introduces an open-source deep learning framework that integrates multi-source satellite imagery and urban geospatial data to detect, map, and analyse UHIs with high spatial fidelity. The framework leverages a U-Net convolutional architecture with attention mechanisms to predict land surface temperature (LST) and delineate UHI hotspots. Input features include NDVI, impervious surface area, building density, and land use classifications, processed through a reproducible pipeline built with open-source tools such as QGIS, TensorFlow, and GDAL. Applied to Lagos, Nigeria, a rapidly urbanizing tropical megacity, the model achieved high predictive performance, successfully identifying critical hot zones and spatial correlations with urban morphology. The results reveal strong associations between UHI intensity and impervious surfaces and inverse correlation with vegetation. The framework’s open architecture, combined with publicly released datasets and modular code, ensures adaptability for use in both data-rich and resource-limited settings. This research contributes a transparent, scalable, and participatory approach to UHI detection, offering actionable insights for climate adaptation, heat risk mitigation, and sustainable urban planning. It underscores the importance of open geospatial AI tools in promoting equitable and data-driven environmental governance.